# Perspectives from a Comprehensive Evaluation of Reconstruction-based Anomaly Detection in ICS

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Electrical & Computer ENGINEERING

S3D

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Carnegie Mellon University

# Industrial control systems (ICS) govern vital infrastructure













https://samcotech.com/common-industrial-water-treatment-problems-how-to-fix-them/
https://www.britannica.com/technology/chemical-industry/Heavy-inorganic-chemicals
https://itrust.sutd.edu.sg/testbeds/secure-water-treatment-swat/
https://lunesys.com/ukrainian-power-grid-hacked/
https://www.pbs.org/wgbh/nova/article/cyber-attack-german-steel-mill-leads-massive-real-world-damage/

# The importance of ICS security is increasing

German Steel Mill (2014)



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COMPUTING

**Triton is the world's most murderous malware, and it's spreading**

The rogue code can disable safety systems designed to prevent catastrophic industrial accidents. It was discovered in the Middle East, but the hackers behind it are now targeting companies in North America and other parts of the world, too.

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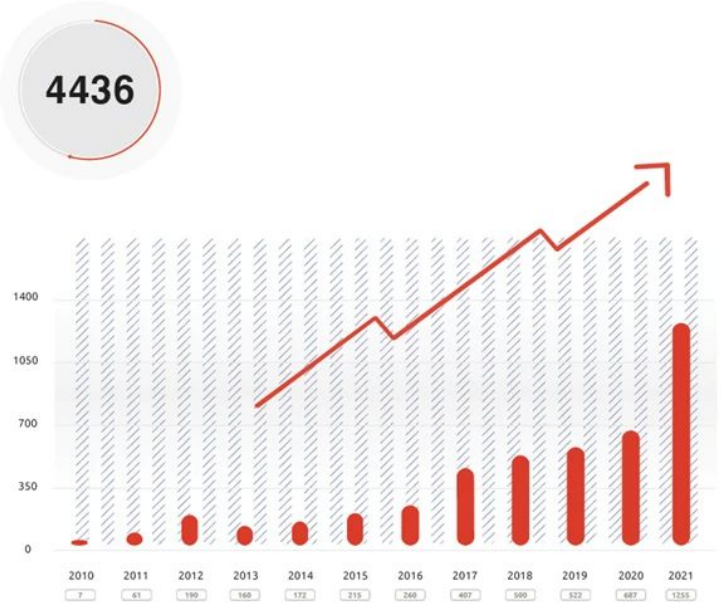
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ICS-CERT CVEs (by year)




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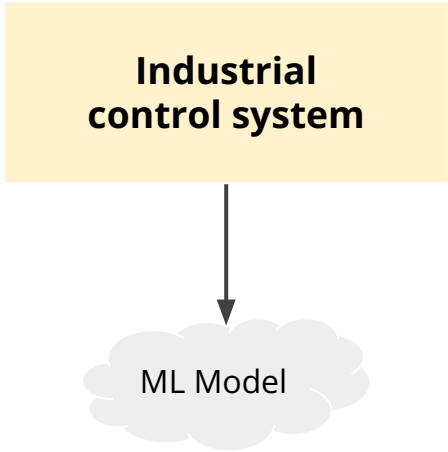
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# ML-based ICS anomaly detection

**Industrial control system**

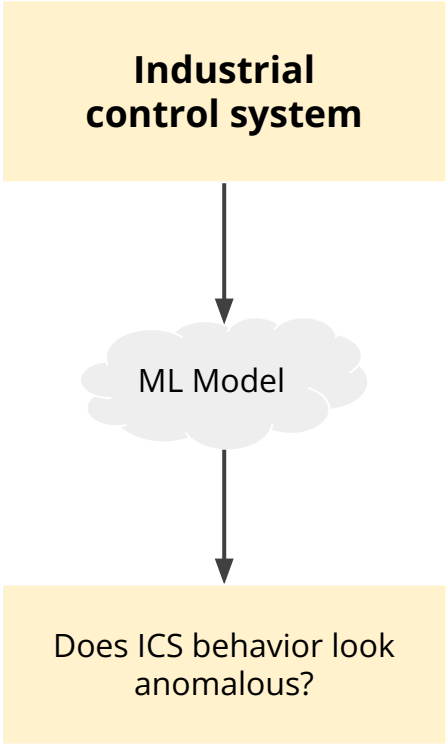
# ML-based ICS anomaly detection

- Learn a model of ICS behavior



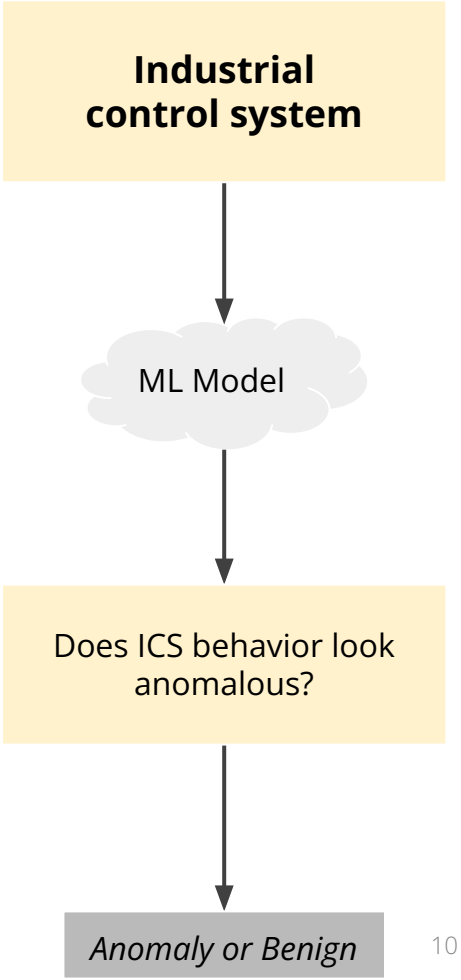
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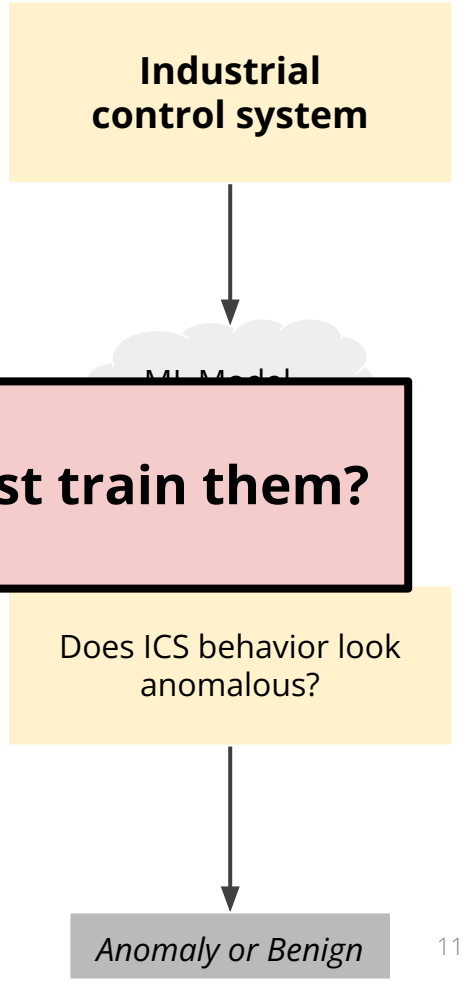
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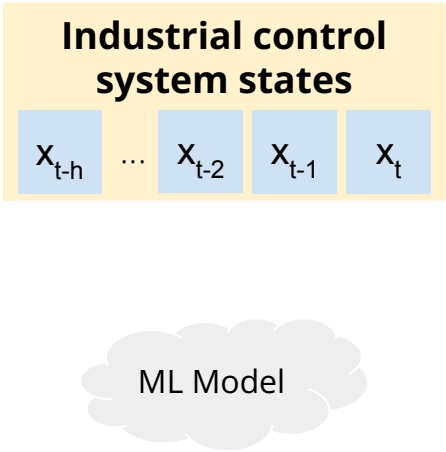
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**We ask: What are the best models and how to best train them?**

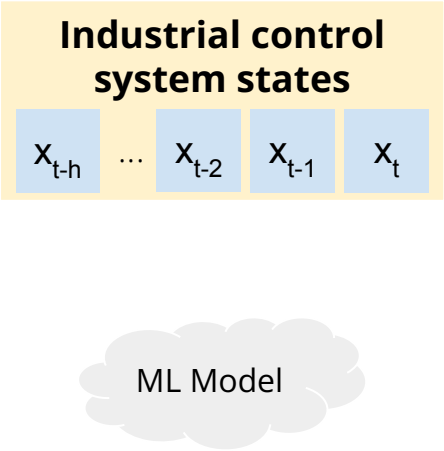
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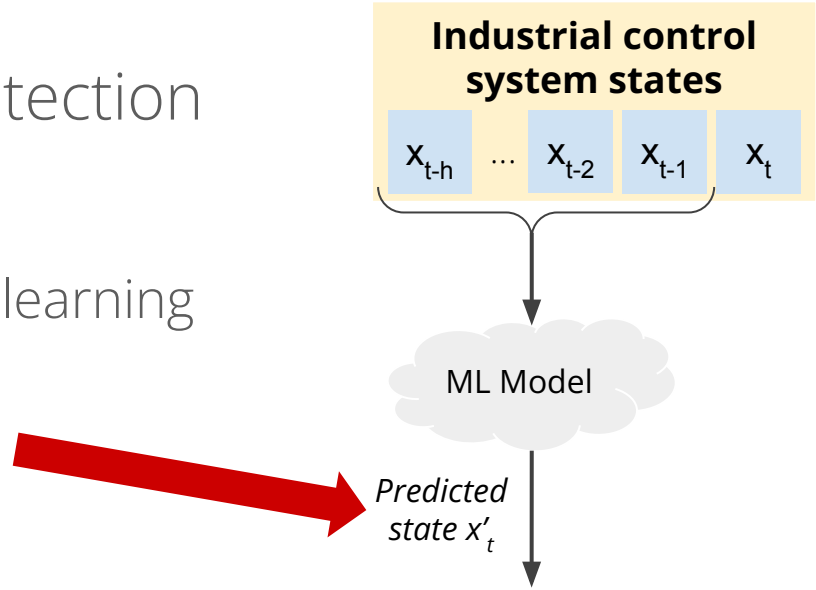
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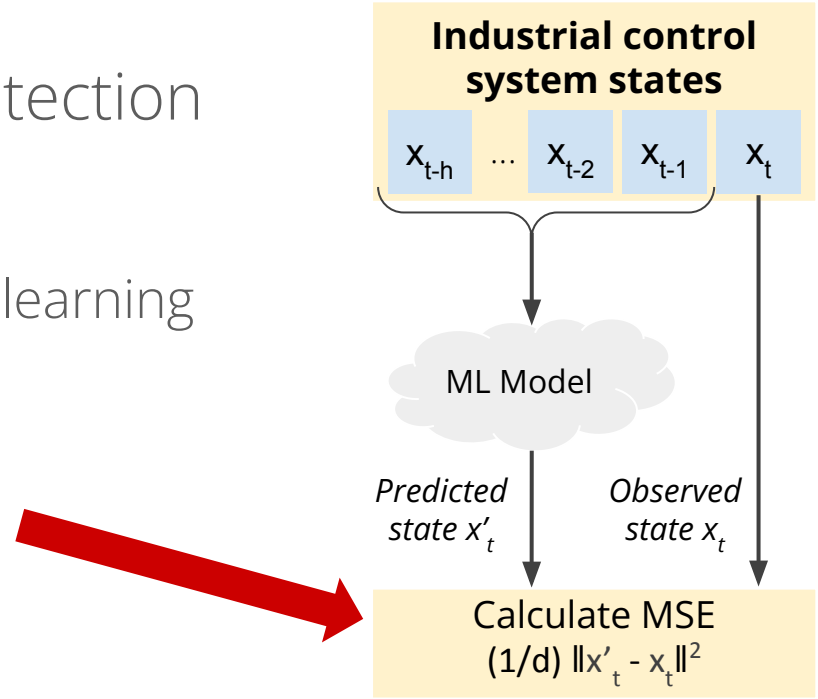
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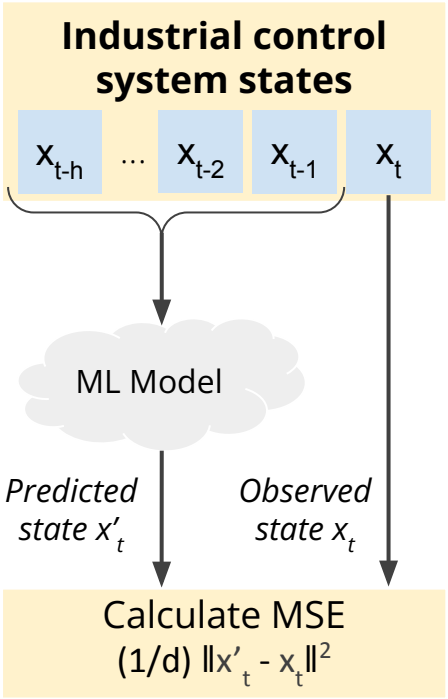
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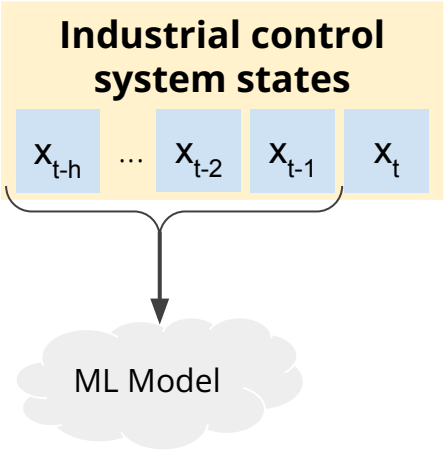
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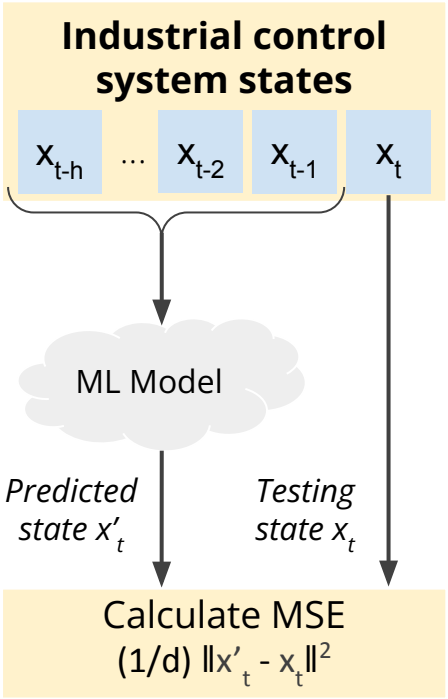
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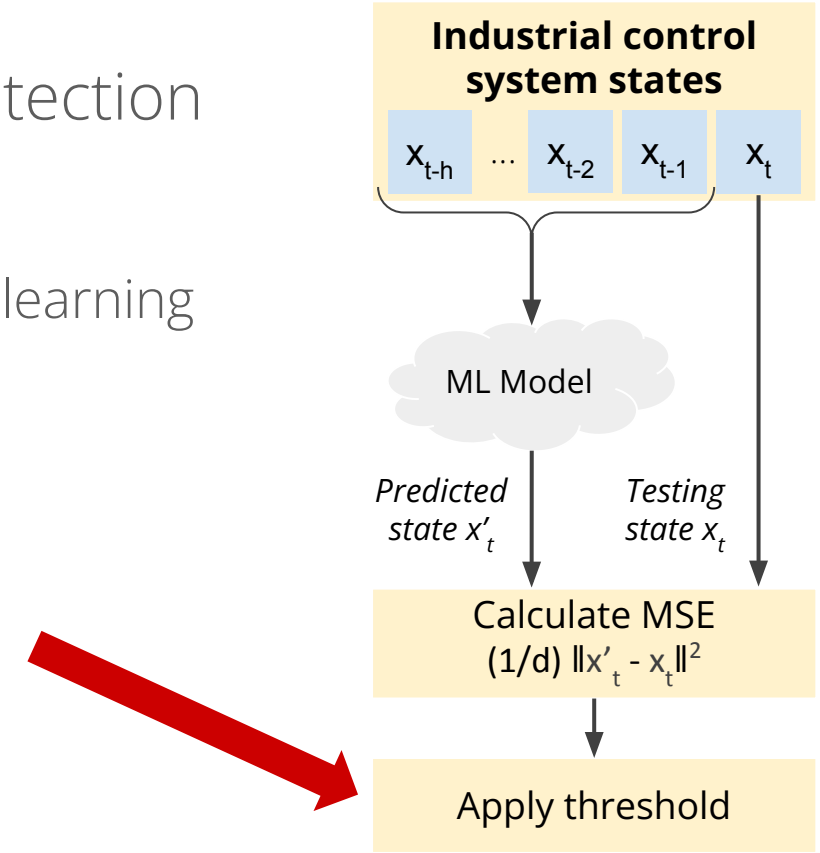
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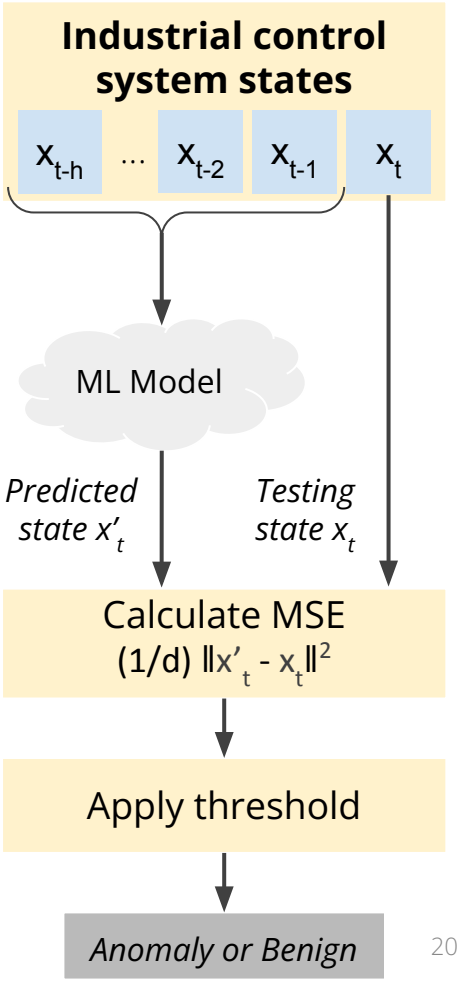
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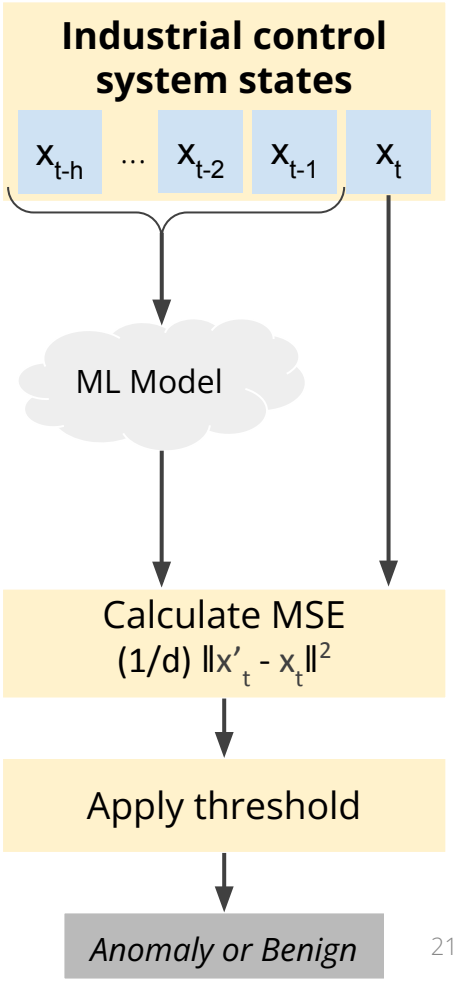
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- At test time:
  - Apply MSE threshold
  - Raise alarms if exceeded



# Prior work uses different…

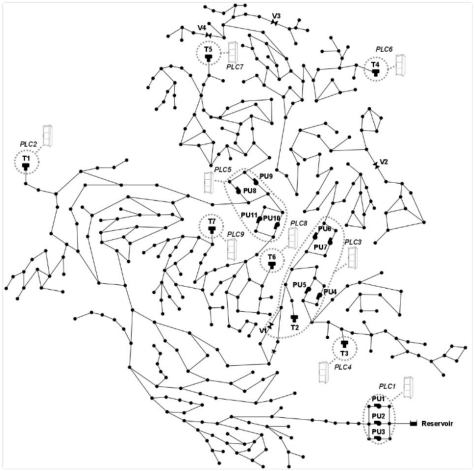
- Datasets
- Architectures
- Techniques

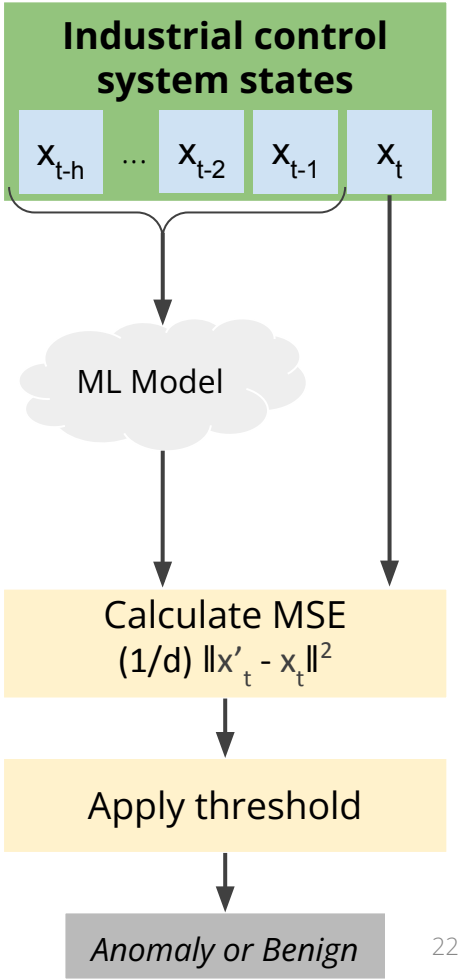


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- **Datasets**:
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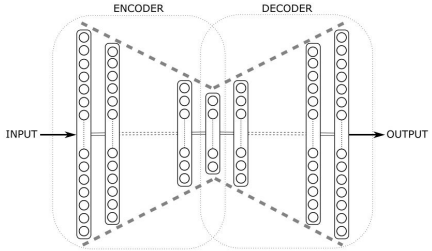
[1] Goh, J., Adepu, S., Junejo, K.N., Mathur, A.: A dataset to support research in the design of secure water treatment systems. *International Conference on Critical Information Infrastructures Security.* 2016.

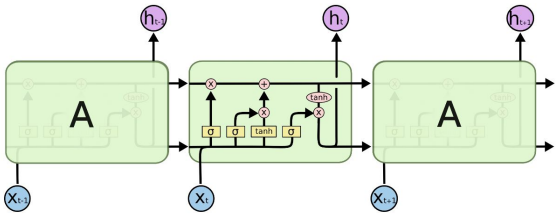
[2] Ahmed, C.M., Palleti, V.R., Mathur, A.: WADI: a water distribution testbed for research in the design of secure cyber physical systems. *3rd International Workshop on Cyber-Physical Systems for Smart Water Networks.* 2017.

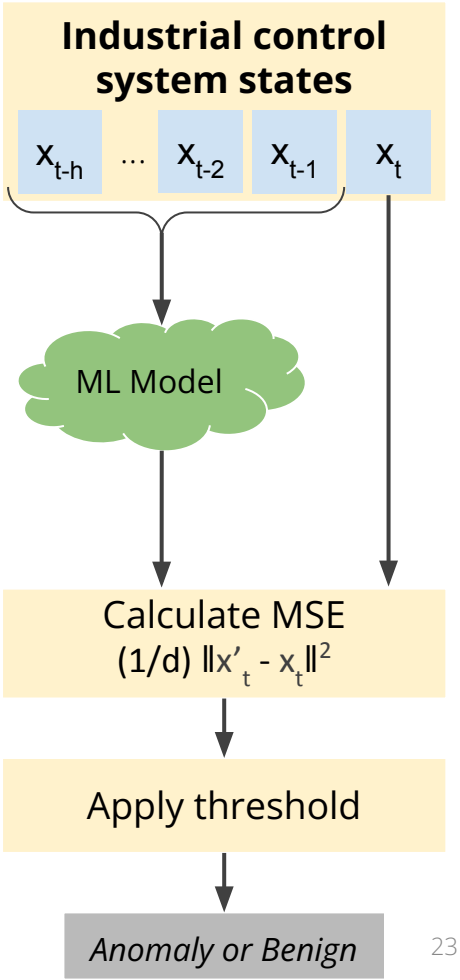
[3] Taormina et al. Battle of the attack detection algorithms: Disclosing cyber attacks on water distribution networks. *Journal of Water Resources Planning and Management 144(8).* 2018.

# Prior work uses different…

- Datasets:
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- **Architectures**:
  - Autoencoders, CNNs, LSTMs



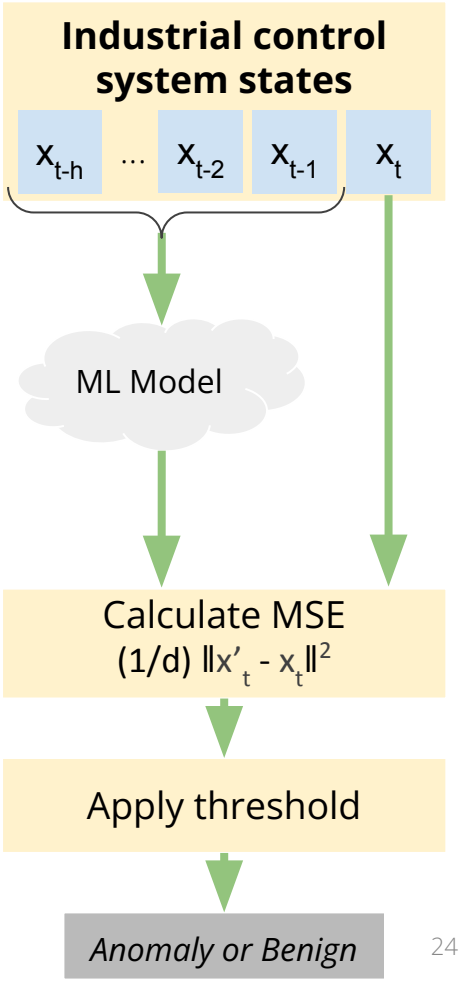




[1] Taormina et al. Deep-Learning Approach to the Detection and Localization of Cyber-Physical Attacks on Water Distribution Systems. *Journal of Water Resources Planning and Management, 144(10).* 2018.
[2] Kravchik et al. Detecting Cyber Attacks in Industrial Control Systems Using Convolutional Neural Networks. CPS-SPC 2018.
[3] Zizzo et al. Intrusion Detection for Industrial Control Systems: Evaluation Analysis and Adversarial Attacks. arXiv 2019.
Images: https://colah.github.io/posts/2015-08-Understanding-LSTMs/

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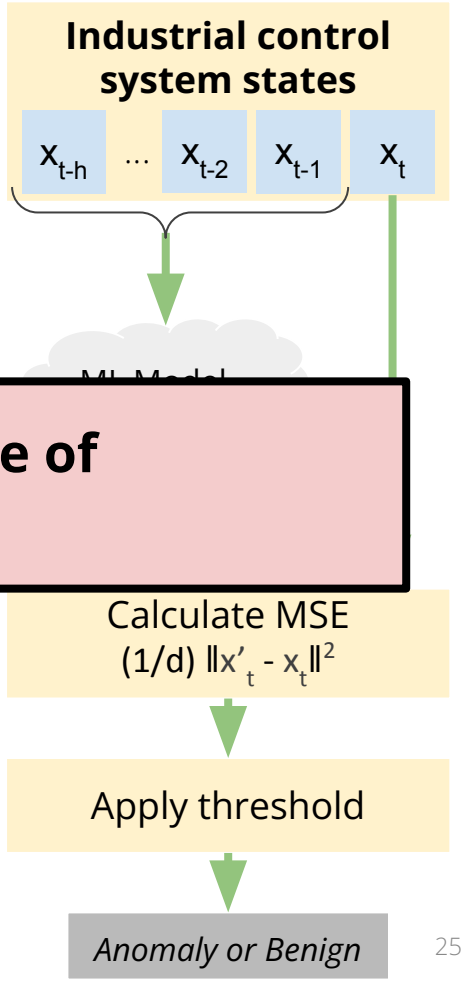
- Datasets:
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- Architectures:
  - Autoencoders, CNNs, LSTMs
- **Techniques**:
  - Early stopping, feature cleaning
  - Various model hyperparameters
  - Various thresholding values



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**Difficult to compare prior work because of inconsistent methodology!**



# What ML architecture is best?

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- We evaluate proposed ICS anomaly detection approaches:
  - Across datasets, model architectures, and hyperparameters
  - With a **common methodology**

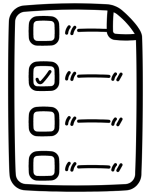


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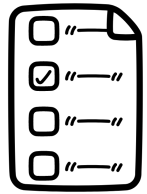


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- We describe the need for different ICS anomaly-detection metrics
  - Explain why we should **stop using the point-F1 score**
  - **Use range-based metrics** for better tuning and optimization



Part 1

# What **models are best** for ICS anomaly detection?

# A common training and evaluation methodology

**1** **Pre-process ICS dataset**

Datasets: SWaT, WADI, BATADAL

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**4** **Evaluate against attacks at test time**

Report final point-F1 score, averaged over 3x random seeds

# A common training and evaluation methodology

**1**

## Pre-process ICS dataset

Datasets: SWaT, WADI, BATADAL
Key techniques:

- *Benign data shuffling*
- *Feature selection*
- *Attack cleaning*

**2**

## Train unsupervised ML model

CNN, LSTM: 1-5 layers, 4-256 units, 50-200 history
AE: 1-5 layers, 1.5-4.0 compression
Key technique: *Early stopping*

**3**

## Tune threshold

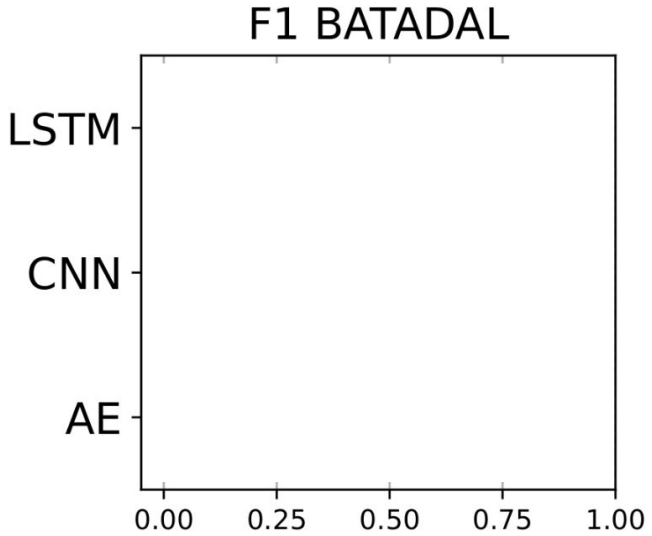
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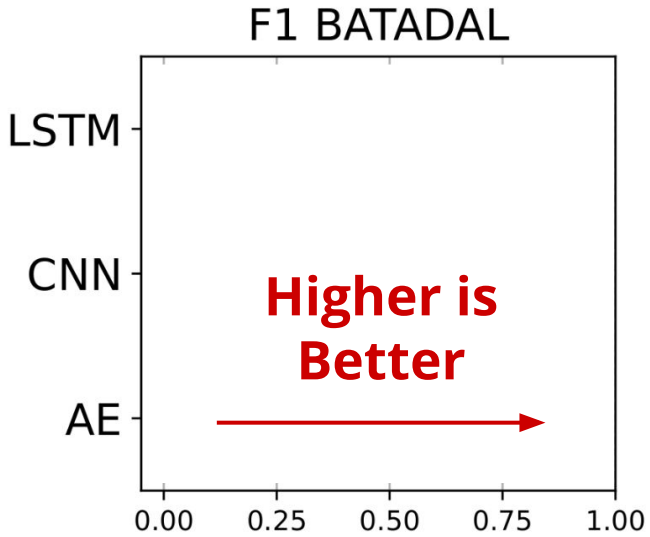
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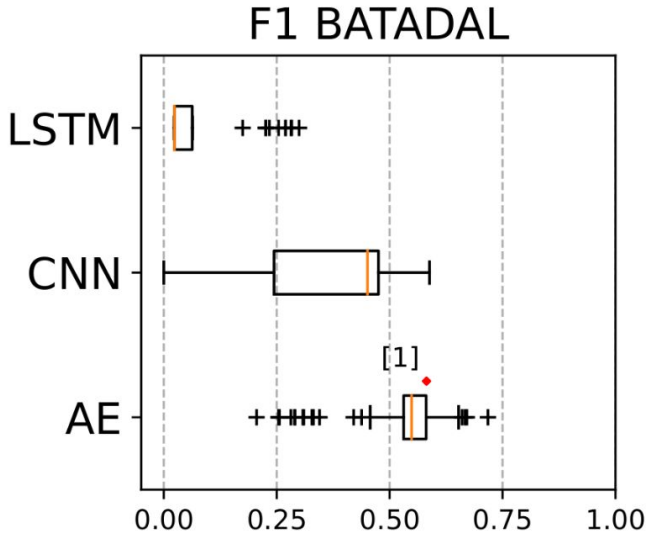
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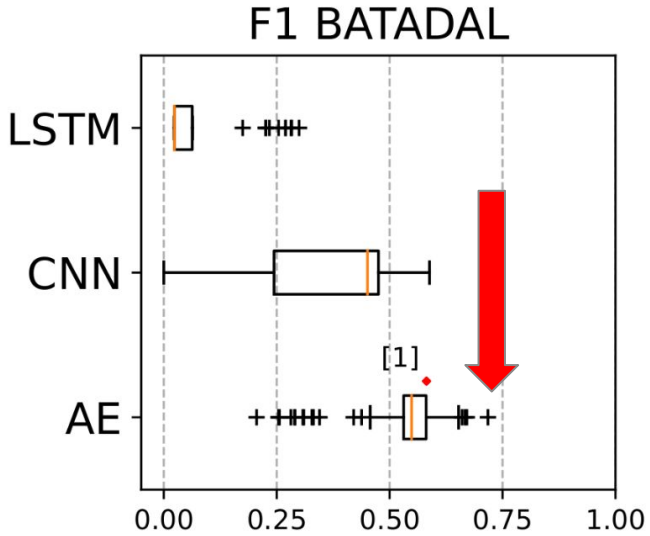
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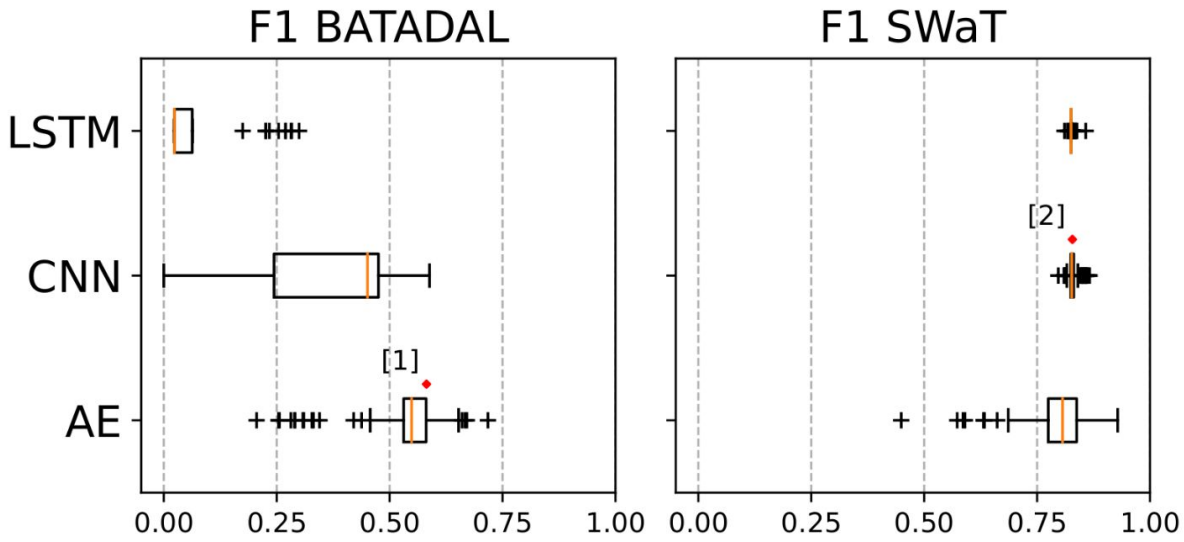
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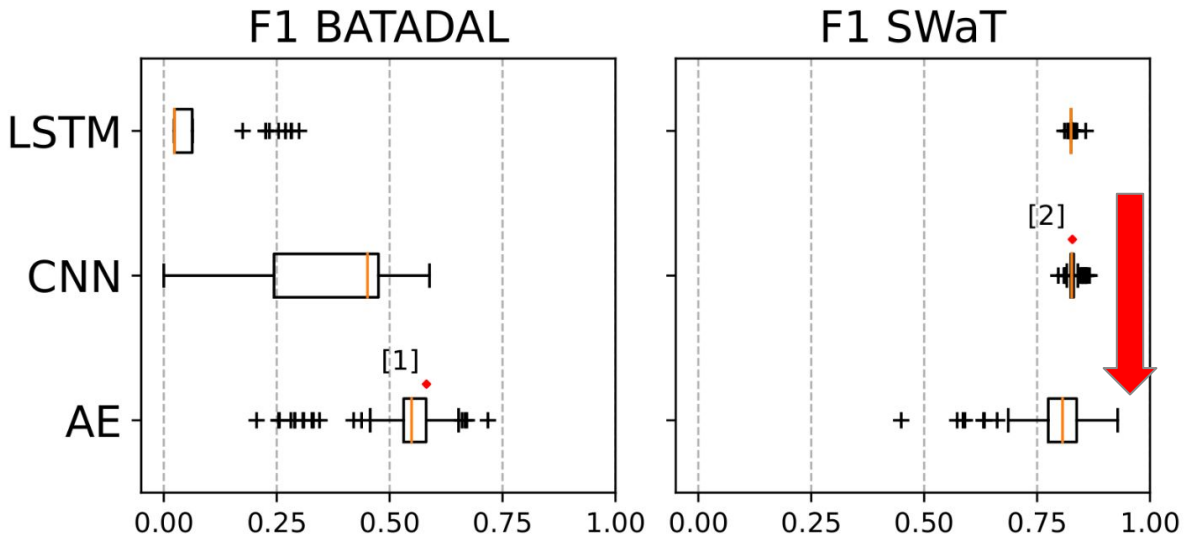
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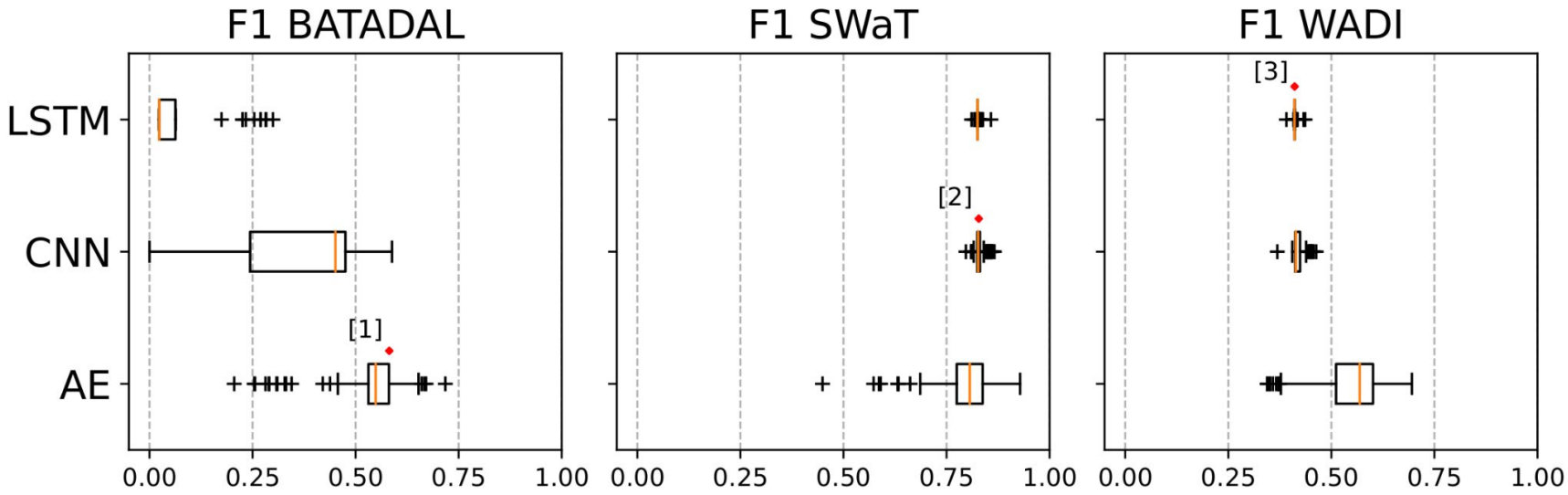
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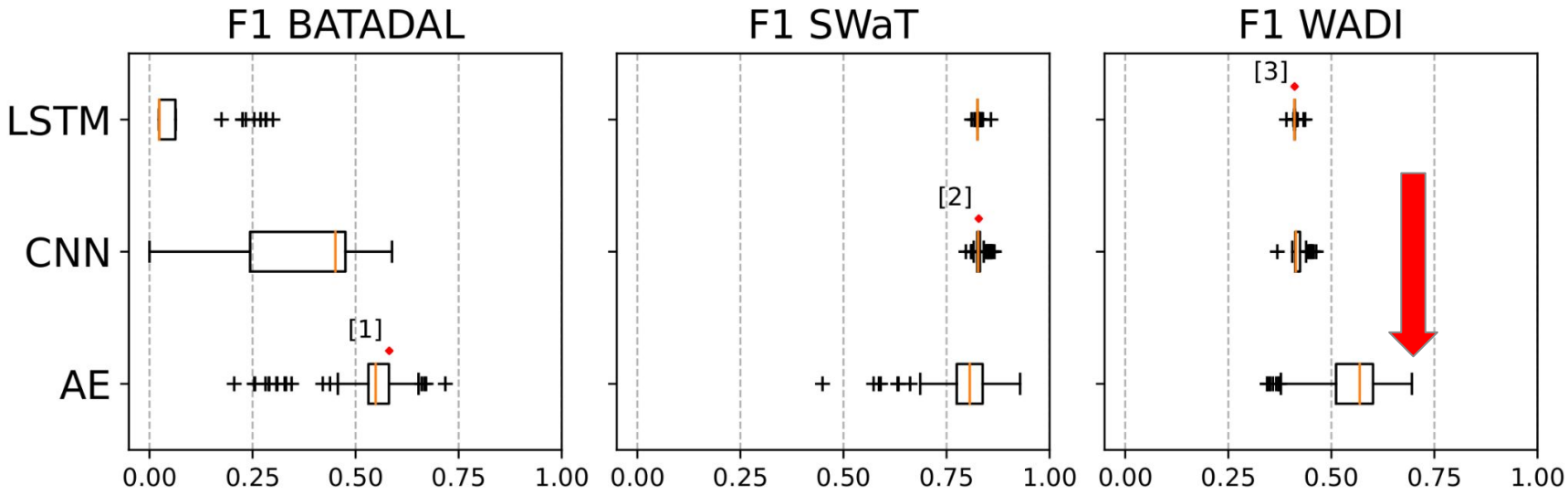
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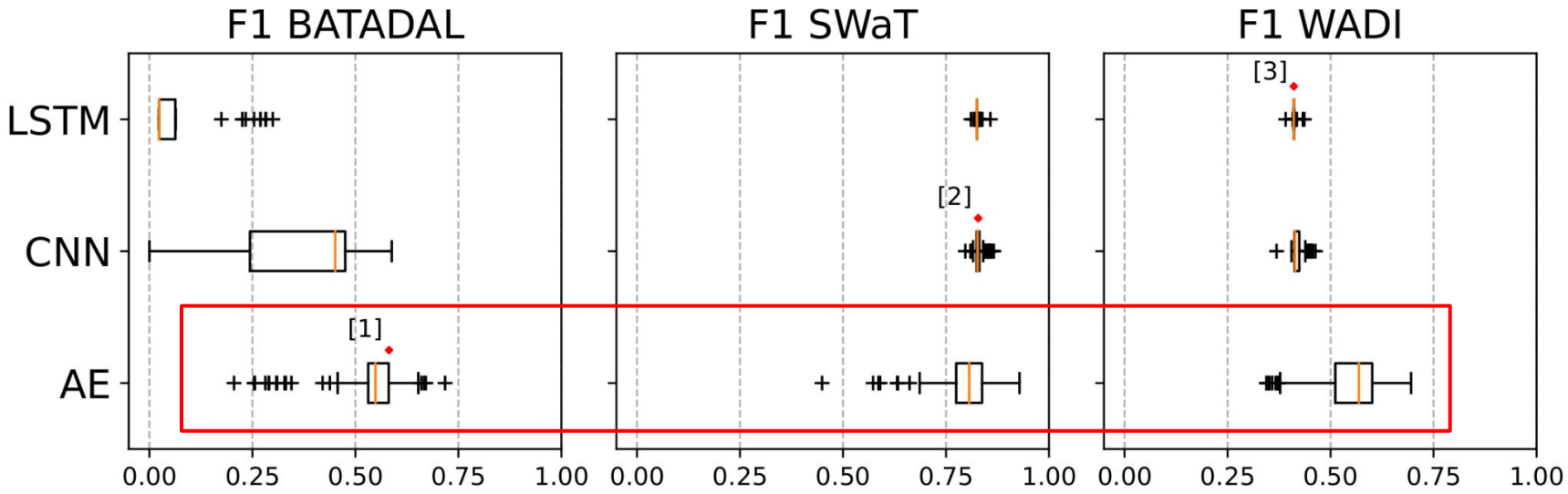
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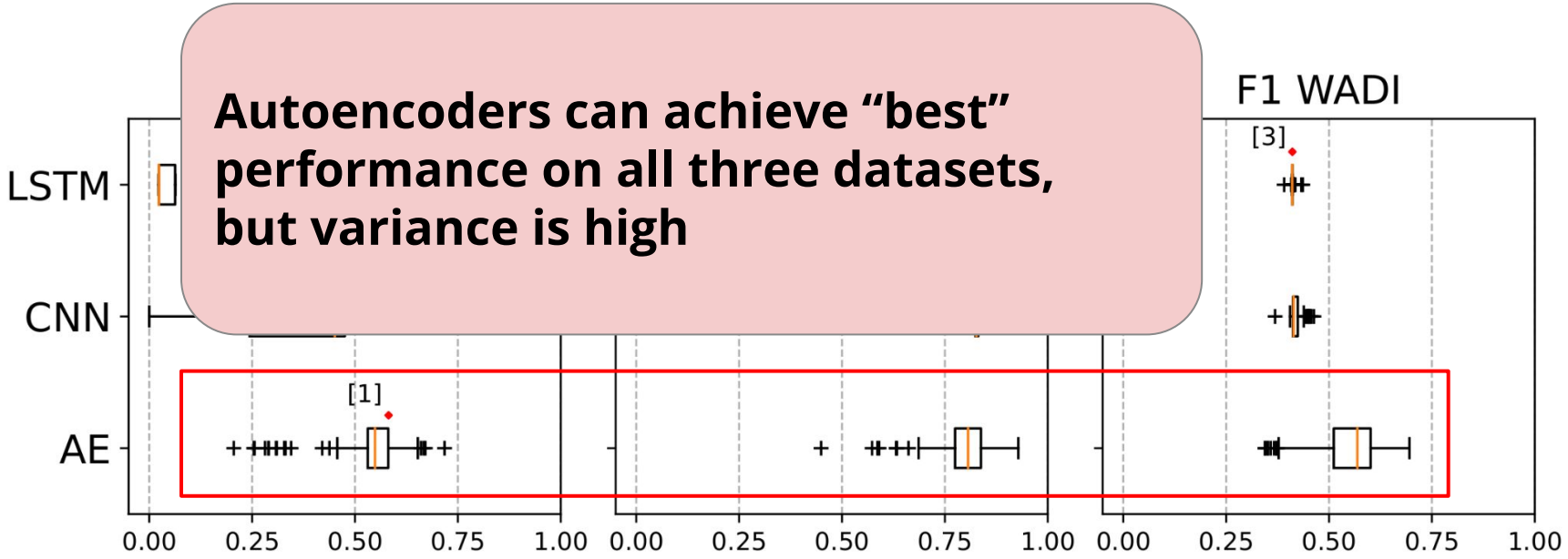
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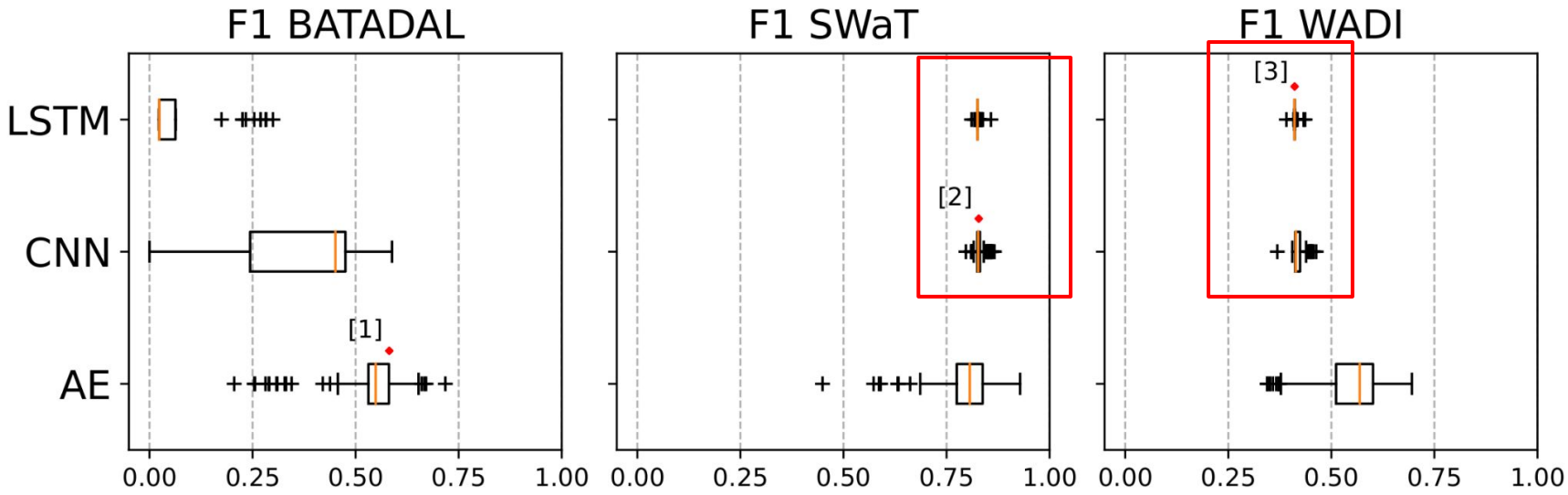
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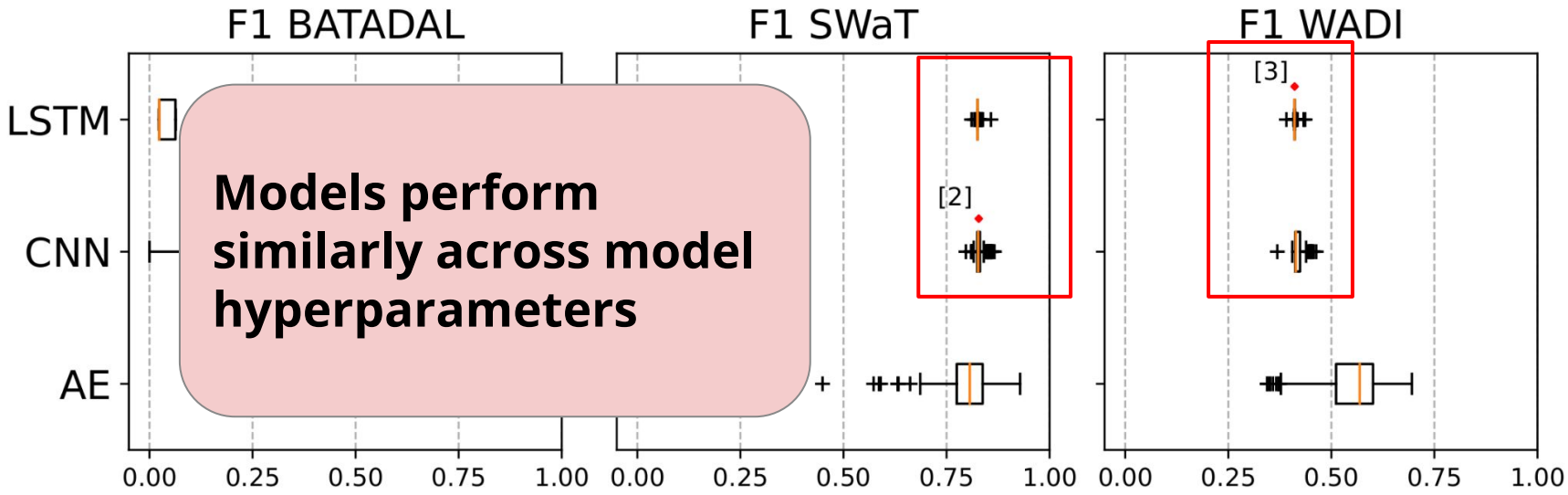
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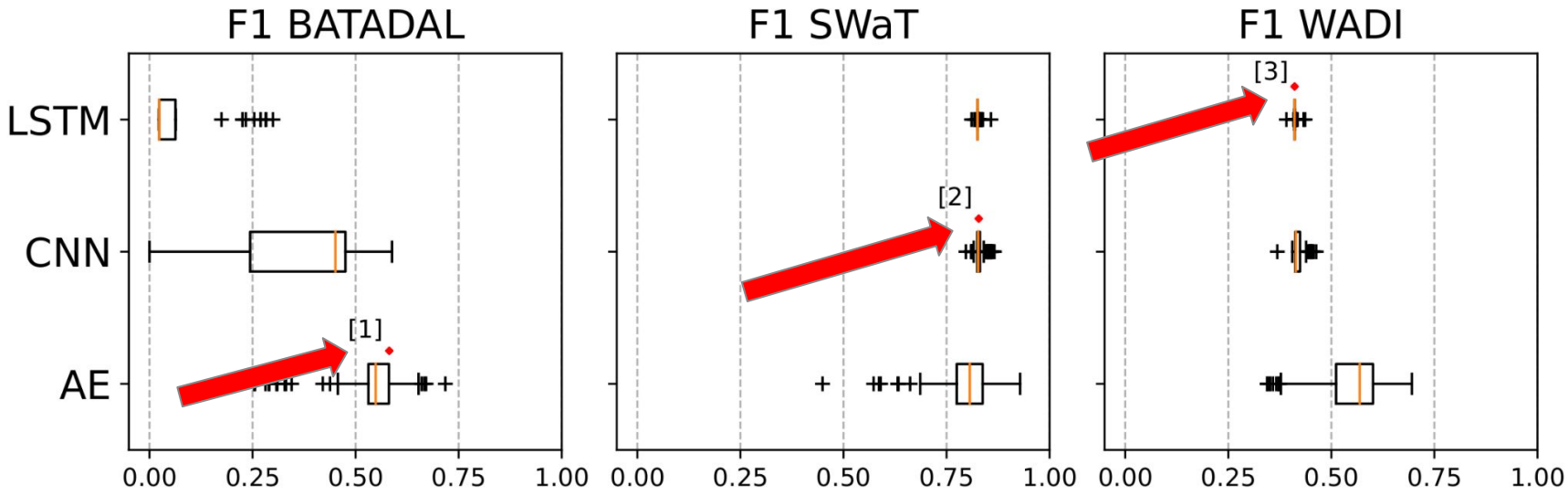
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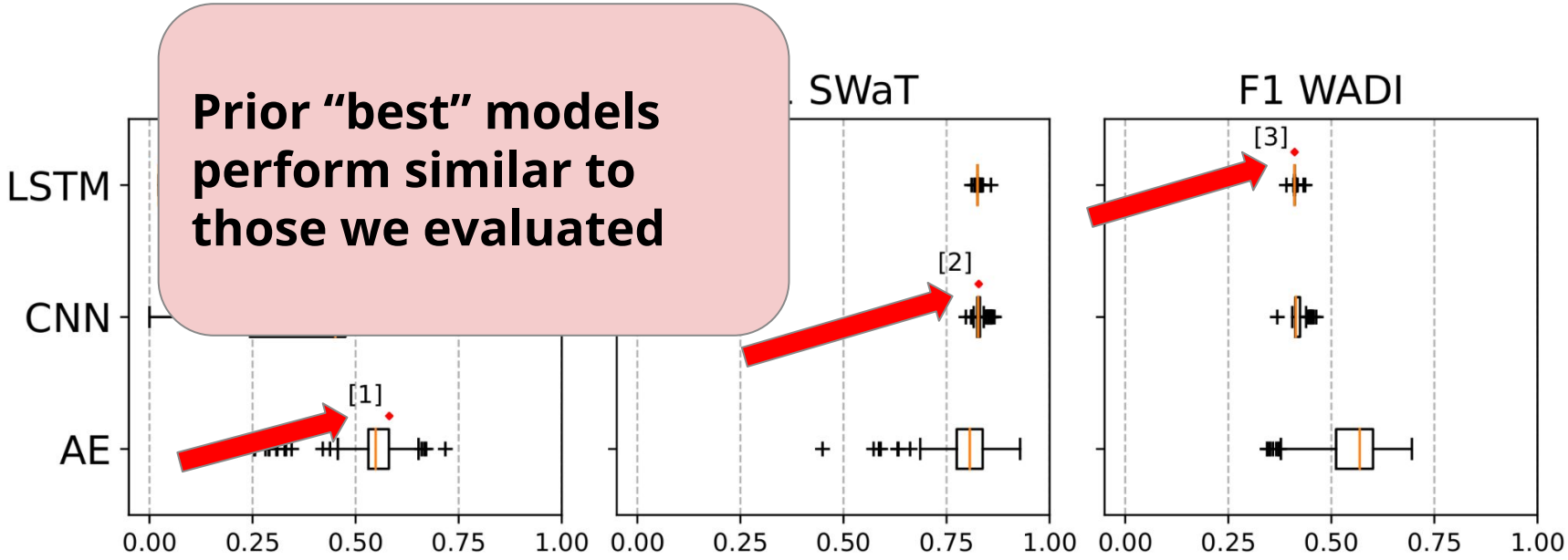
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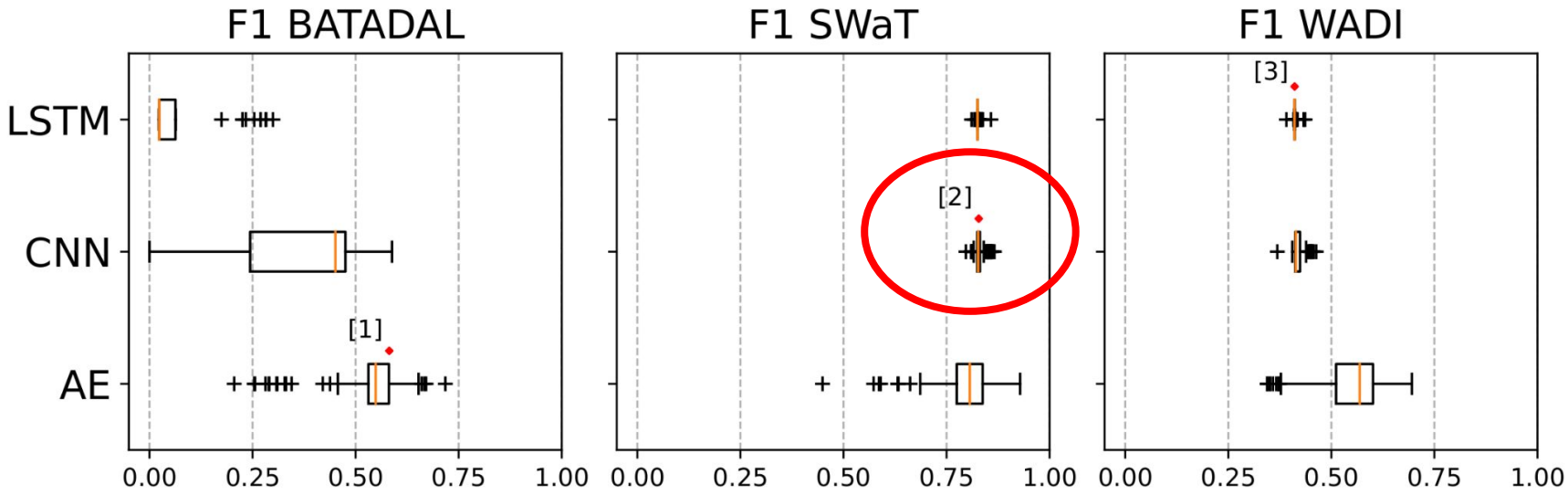
# Is there a best model?



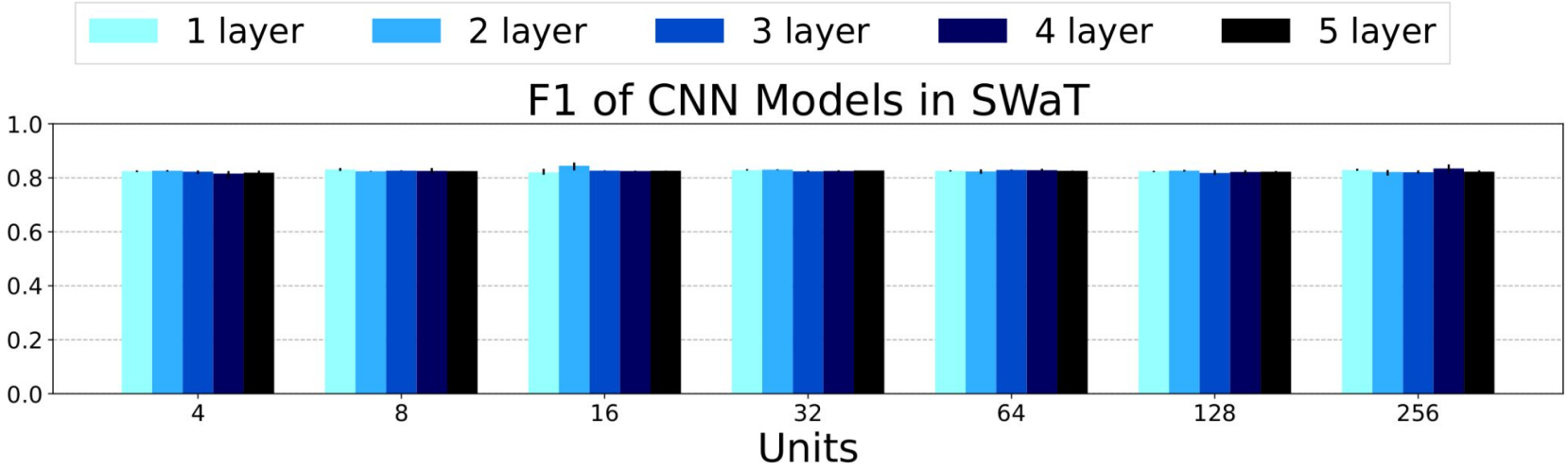
# Is there a best model?



# How important is model hyperparameter tuning?



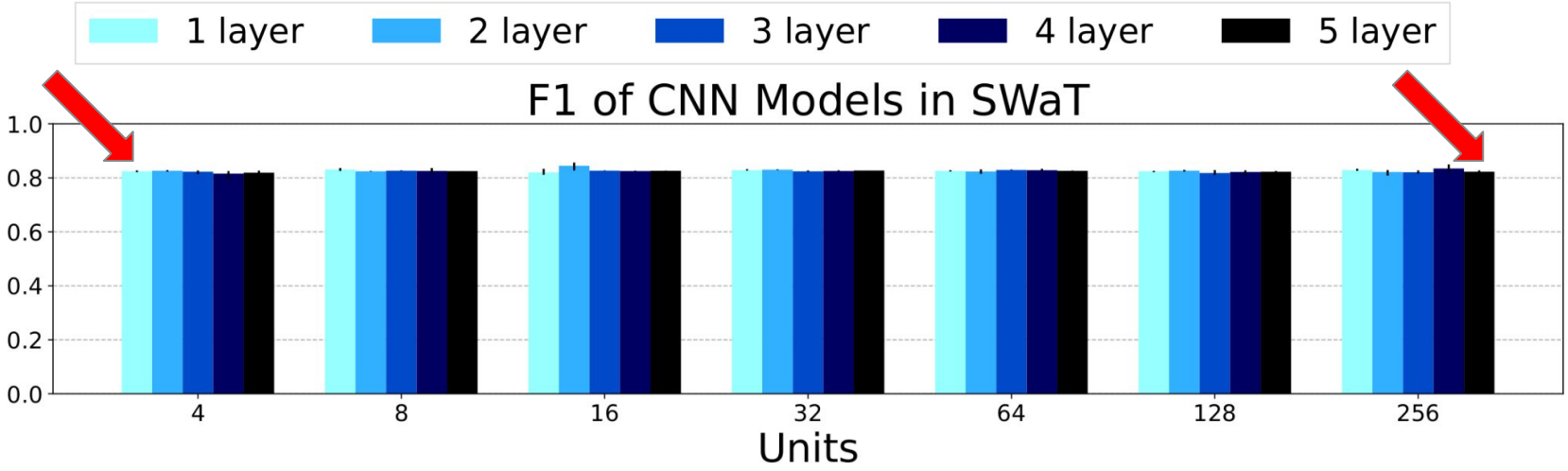
# How important is model hyperparameter tuning?



# How important is model hyperparameter tuning?

Smallest model: F1 = **0.824**

Largest model: F1 = **0.823**

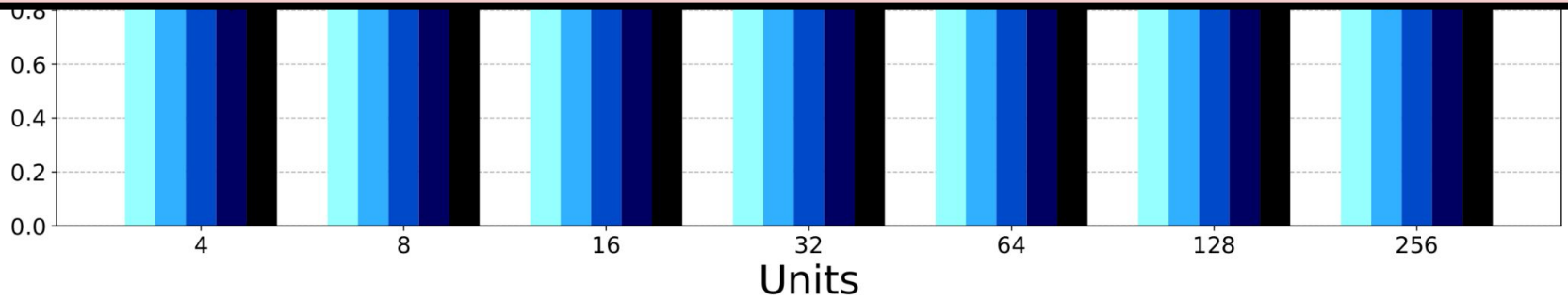


# How important is model hyperparameter tuning?

Smallest model: F1 = **0.824**

Largest model: F1 = **0.823**

**Model hyperparameter tuning has a limited impact on performance**



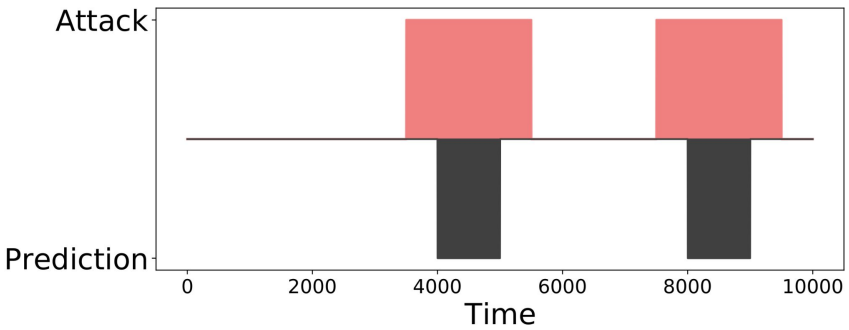
Part 2

# How do **range-based metrics** affect tuning and optimization?

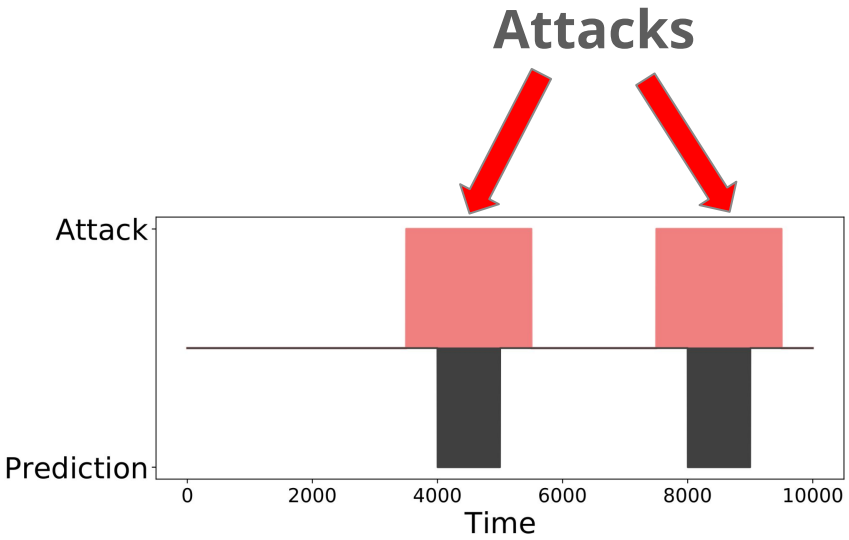
# Point-F1: a common metric in ICS anomaly detection

- Point-F1 = Average between precision and recall
  - Each instance in time is equally weighted
- But attacks and predictions are **performed in segments**

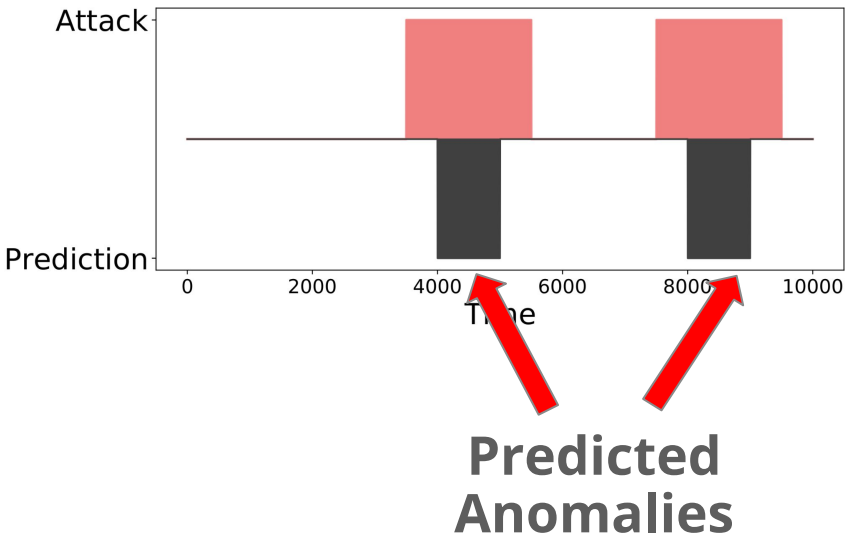
# Point-F1 does not capture segment-based objectives



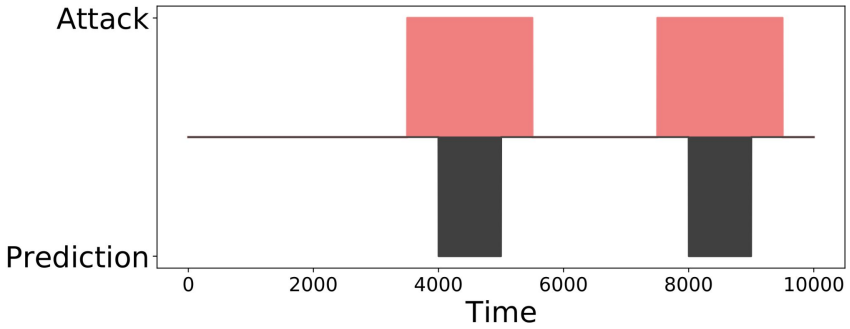
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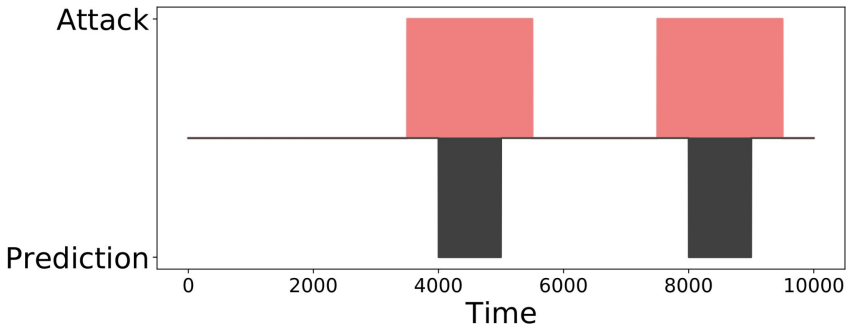
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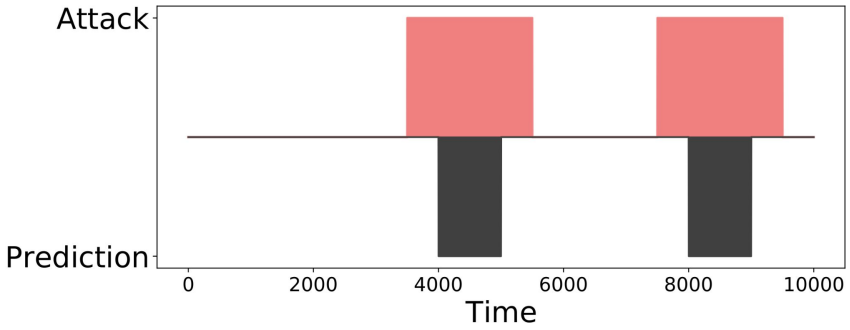


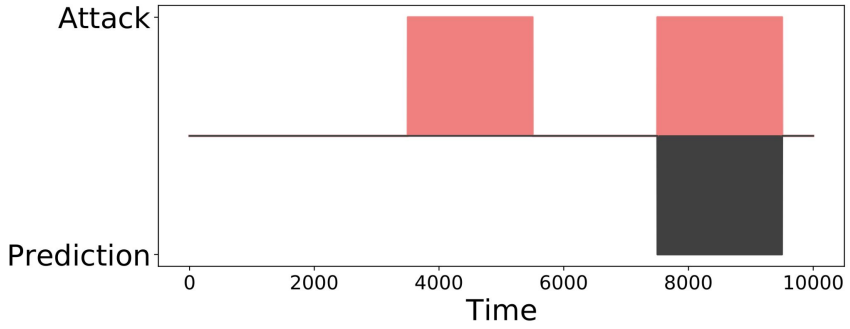
# Point-F1 does not capture segment-based objectives




- Both attacks partially detected

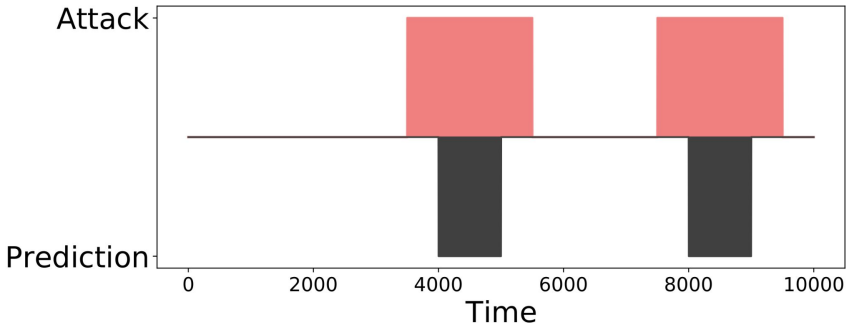
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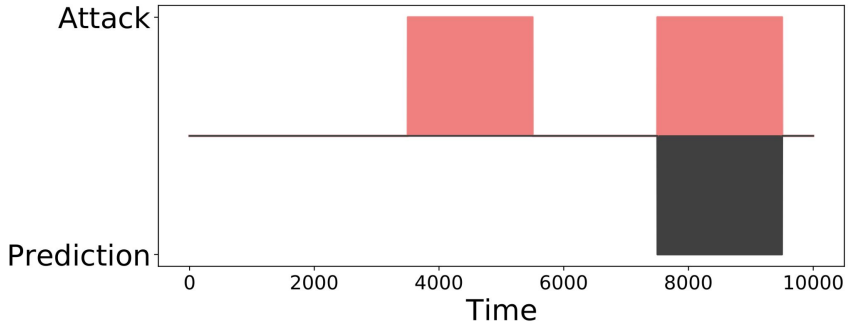




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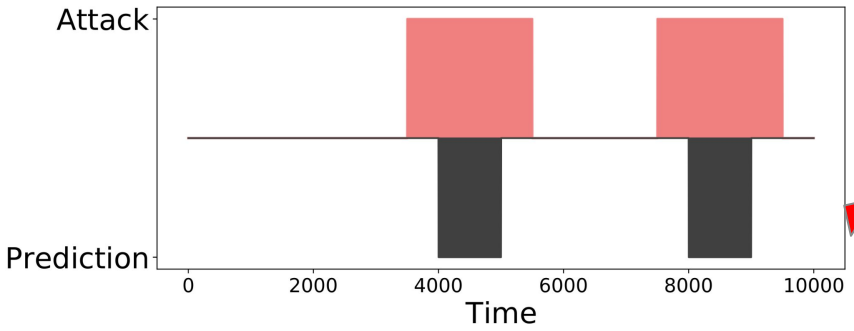



- Both attacks partially detected



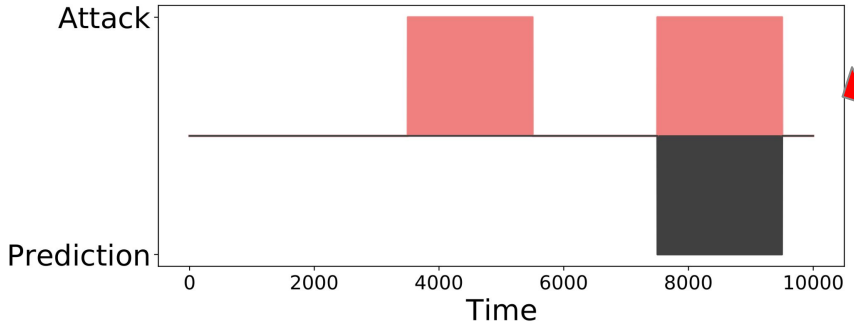

- One attack completely missed
- One attack fully detected

# Point-F1 does not capture segment-based objectives



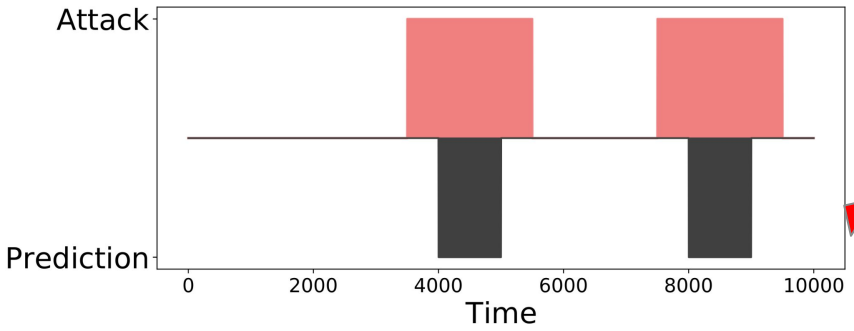

- Both attacks partially detected

Point-F1 = 0.75



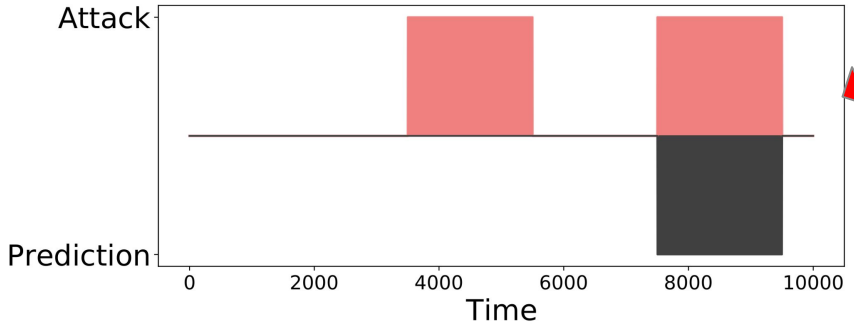

- One attack completely missed
- One attack fully detected

# Point-F1 does not capture segment-based objectives




- Both attacks partially detected

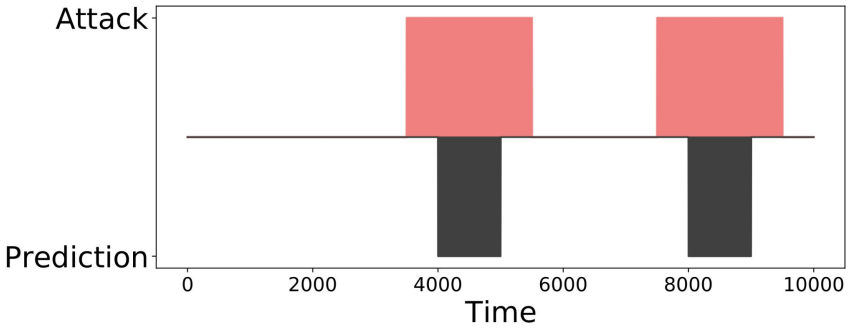
Point-F1 = 0.75




- One attack completely missed
- One attack fully detected

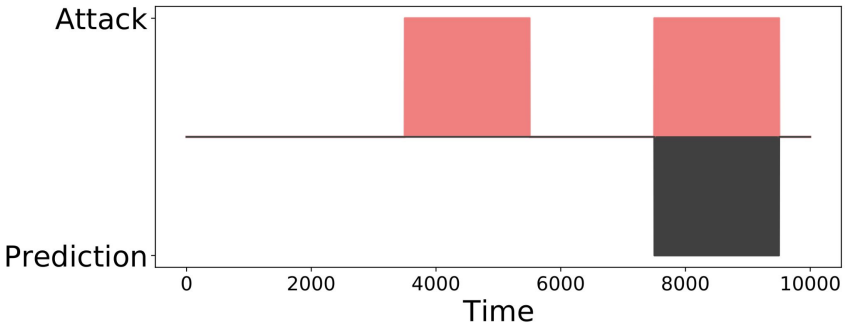
**Same point-F1 score, but different outcomes!**

# Which objectives are important?




**Detect every attack?**

**vs.**

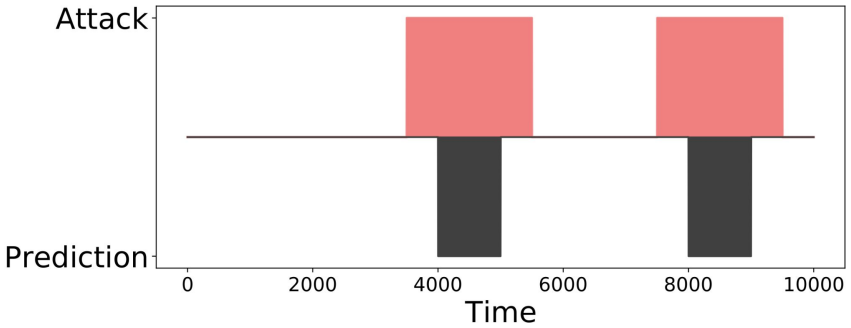



**Detect all of attack?**



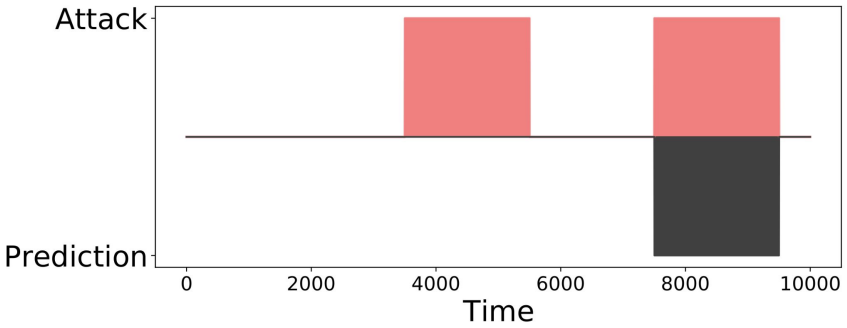
[1] Tatbul, N., Lee, T.J., Zdonik, S., Alam, M., Gottschlich, J.: Precision and recall for time series. *NeurIPS 2018*.

# Which objectives are important?




**Detect every attack?**

**vs.**



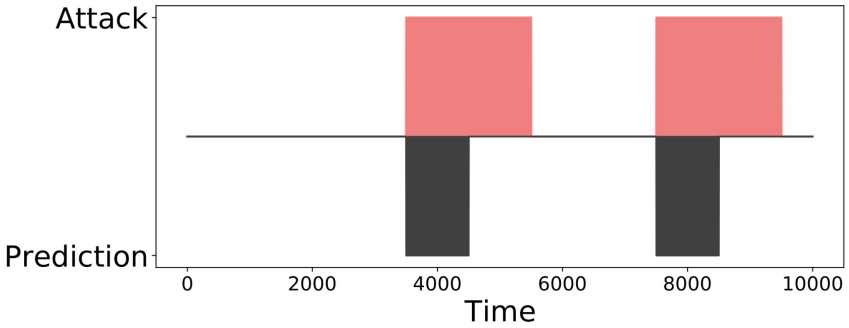

**Detect all of attack?**

- Detected **segments**, instead of detected timesteps
  - Captured by time-aware precision and recall metric [1]



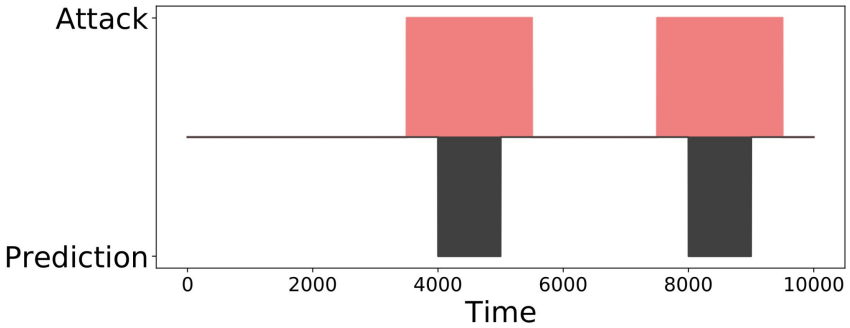
[1] Tatbul, N., Lee, T.J., Zdonik, S., Alam, M., Gottschlich, J.: Precision and recall for time series. *NeurIPS 2018*.

# Which objectives are important?




**Detect immediately?**

**vs.**

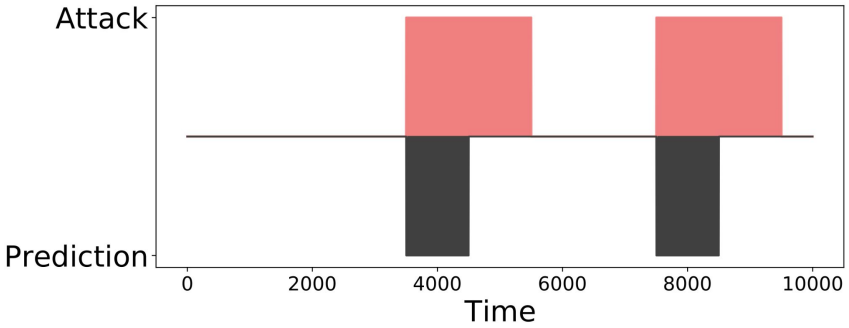



**Detect later?**



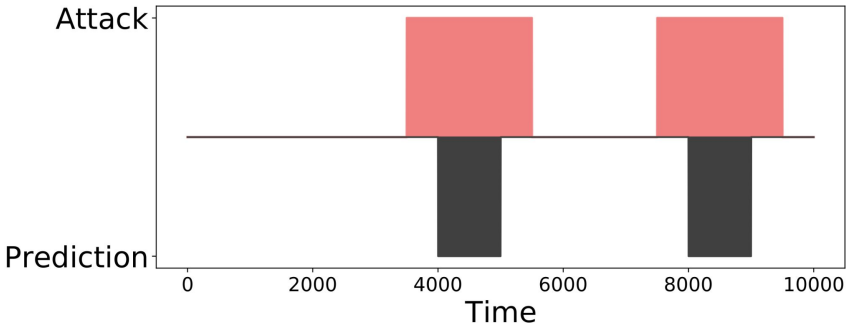
[1] Tatbul, N., Lee, T.J., Zdonik, S., Alam, M., Gottschlich, J.: Precision and recall for time series. *NeurIPS 2018*.
[2] Lavin, A., Ahmad, S.: Evaluating real-time anomaly detection algorithms–the Numenta anomaly benchmark. *14th International Conference on Machine Learning and Applications*. (2015)

# Which objectives are important?




**Detect immediately?**

**vs.**




**Detect later?**

- Detecting attacks **earlier**, rather than later
  - Captured by Numenta metric [2]



[1] Tatbul, N., Lee, T.J., Zdonik, S., Alam, M., Gottschlich, J.: Precision and recall for time series. *NeurIPS 2018*.
[2] Lavin, A., Ahmad, S.: Evaluating real-time anomaly detection algorithms–the Numenta anomaly benchmark. *14th International Conference on Machine Learning and Applications*. (2015)

# New training and evaluation methodology

## 1 Pre-process ICS dataset

Datasets: SWaT, WADI, BATADAL
Key techniques:

- *Benign data shuffling*
- *Feature selection*
- *Attack cleaning*

## 2 Train unsupervised ML model

CNN, LSTM: 1-5 layers, 4-256 units, 50-200 history
AE: 1-5 layers, 1.5-4.0 compression
Key technique: *Early stopping*

## 3 Tune threshold

MSE threshold **τ**, window length **w**
*objective*: maximize ~~point-F1 score~~ range-F1/Numenta scores

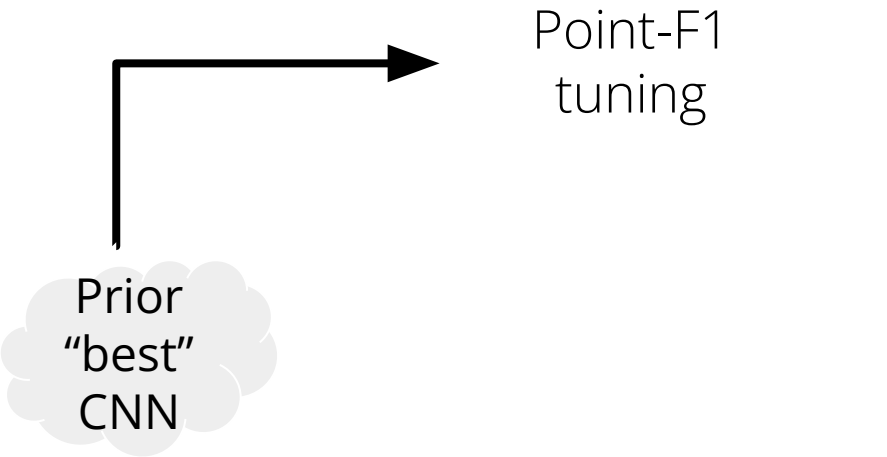
## 4 Evaluate against attacks at test time

Report ~~final point-F1 score~~: attack precision, attack recall, early detection, range-F1

# Flexible tunings with range-based metrics

Prior “best” CNN

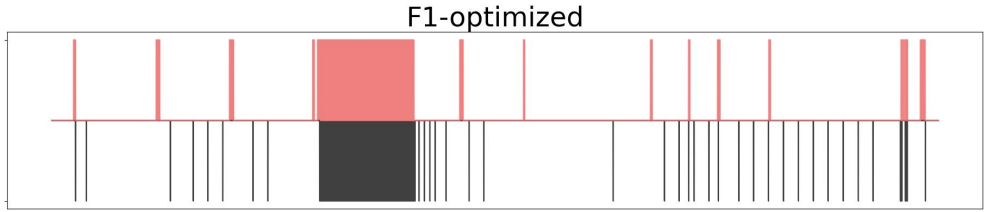
# Flexible tunings with range-based metrics



# Flexible tunings with range-based metrics

Prior "best" CNN

Point-F1 tuning



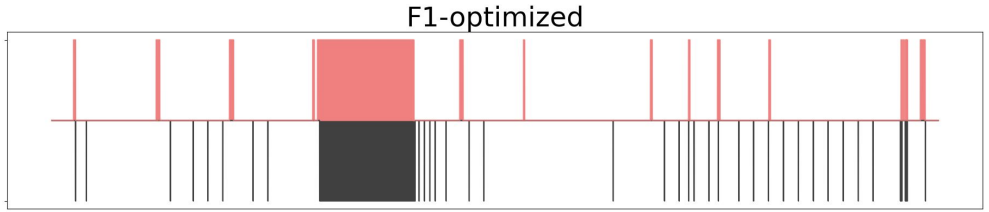

6/18 detected, 32 false alarms

# Flexible tunings with range-based metrics

Prior "best" CNN

Point-F1 tuning

Range-F1 tuning (high precision)



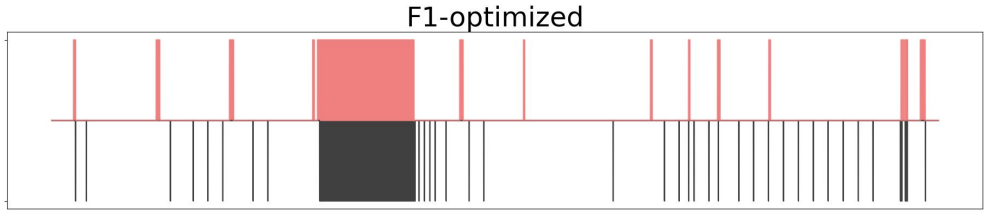

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# Flexible tunings with range-based metrics

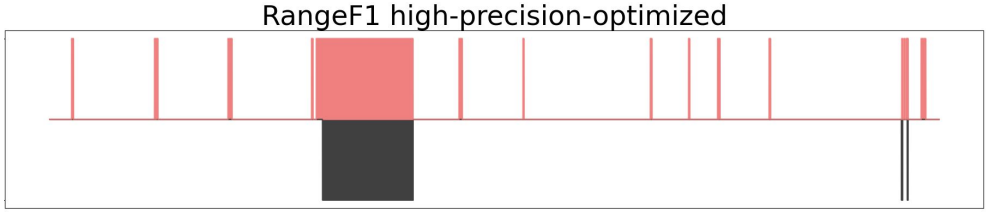
Prior "best" CNN

Point-F1 tuning

Range-F1 tuning (high precision)




6/18 detected, 32 false alarms




3/18 detected, **no false alarms**

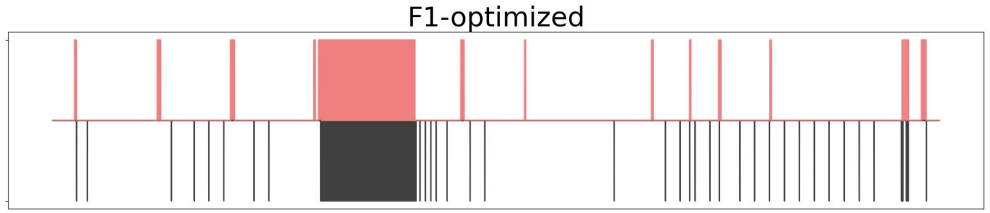
# Flexible tunings with range-based metrics

Prior "best" CNN

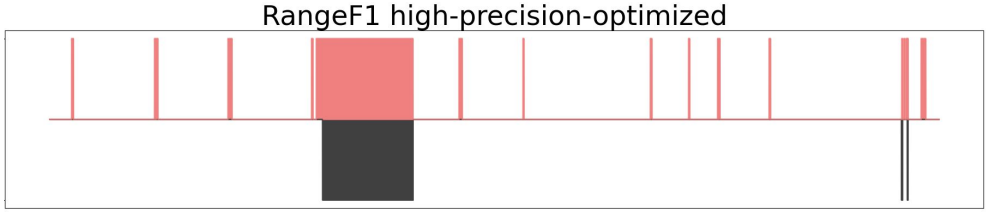
Point-F1 tuning

Range-F1 tuning (high precision)

Numenta tuning (high recall)




6/18 detected, 32 false alarms



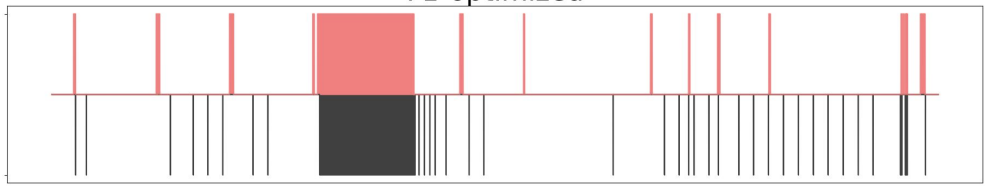

3/18 detected, **no false alarms**

# Flexible tunings with range-based metrics

Prior "best" CNN

Point-F1 tuning

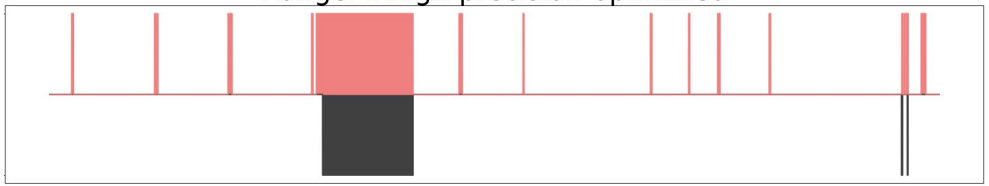
F1-optimized



6/18 detected, 32 false alarms

Range-F1 tuning (high precision)

RangeF1 high-precision-optimized



3/18 detected, **no false alarms**

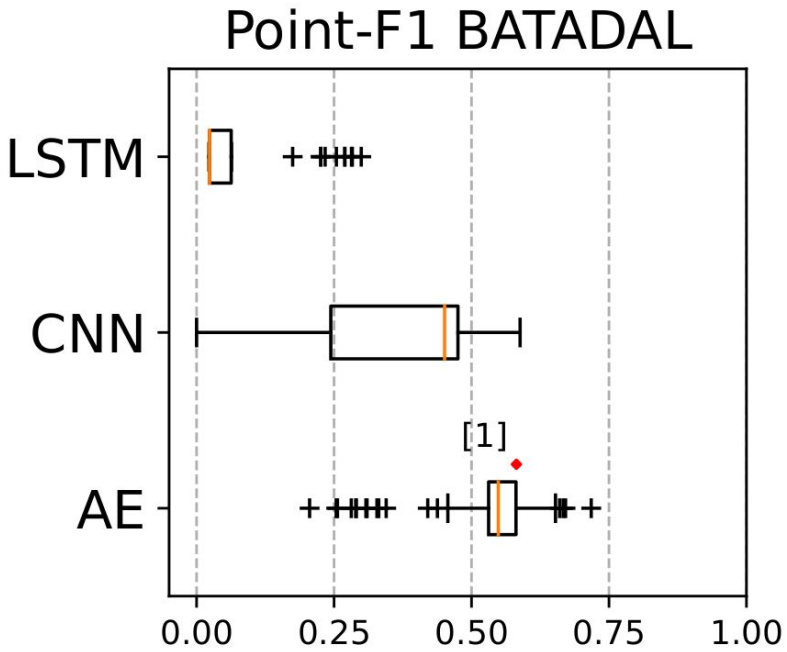
Numenta tuning (high recall)

Numenta high-recall-optimized

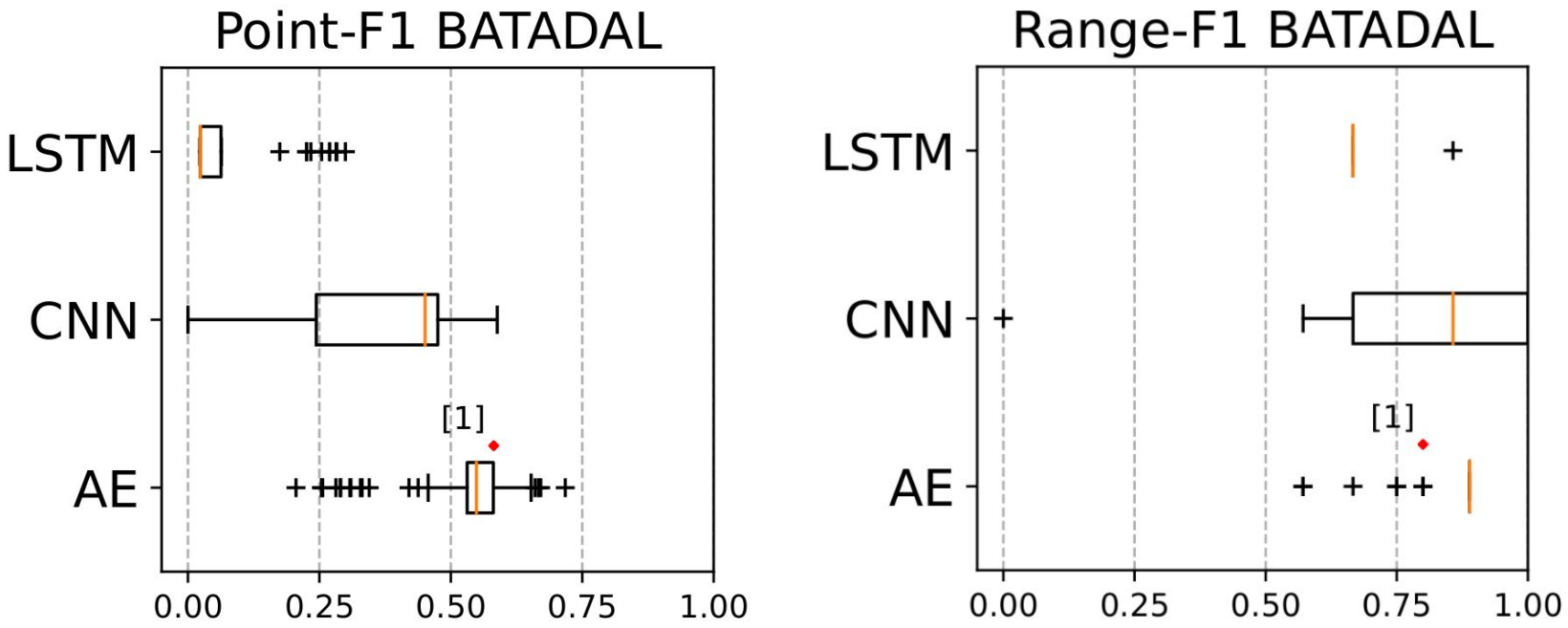


**11/18 detected**, 89 false alarms
7 early detections

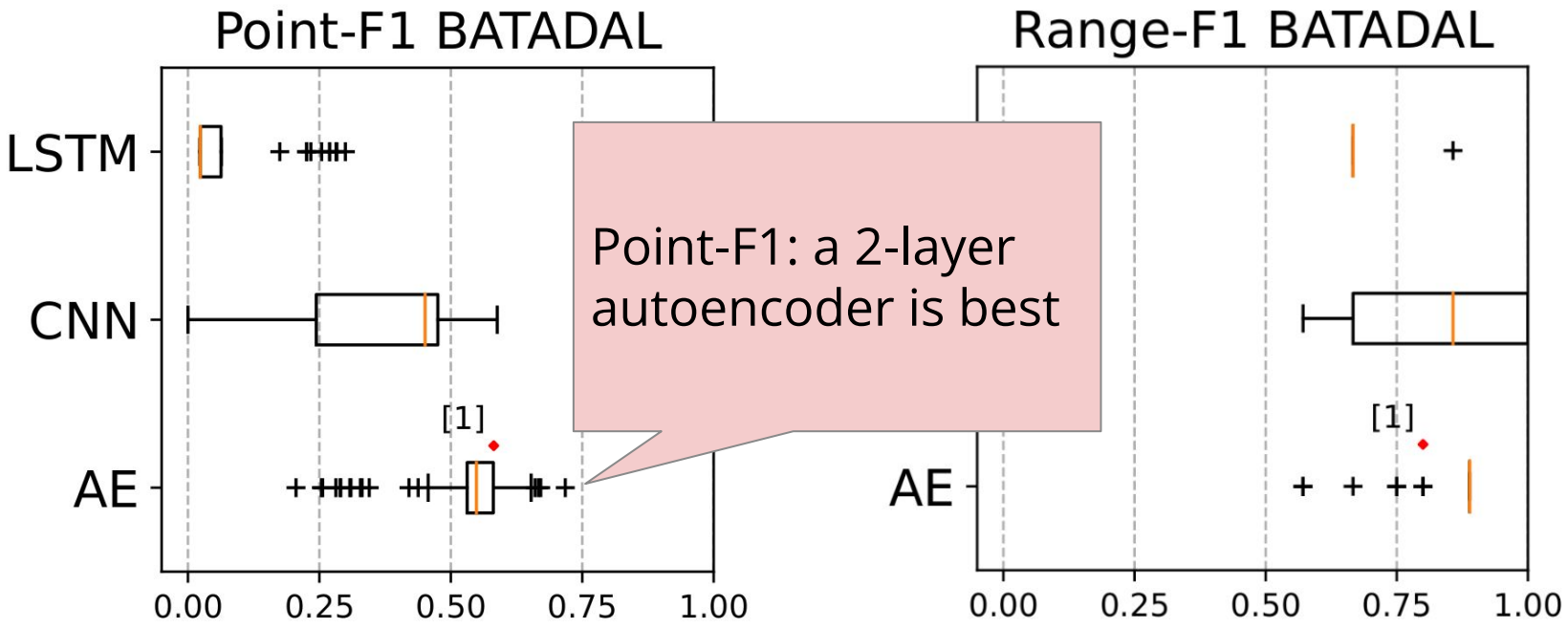
# Range-based metrics choose different optimals



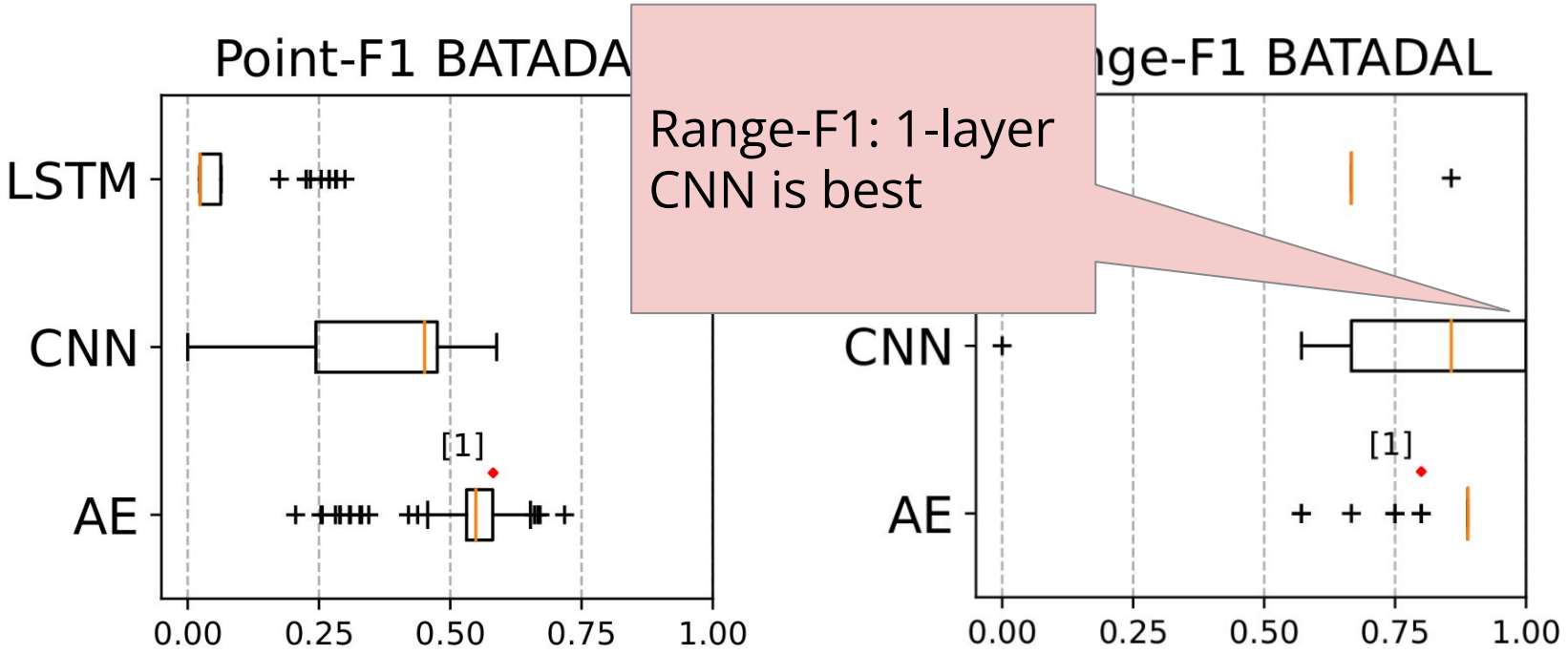
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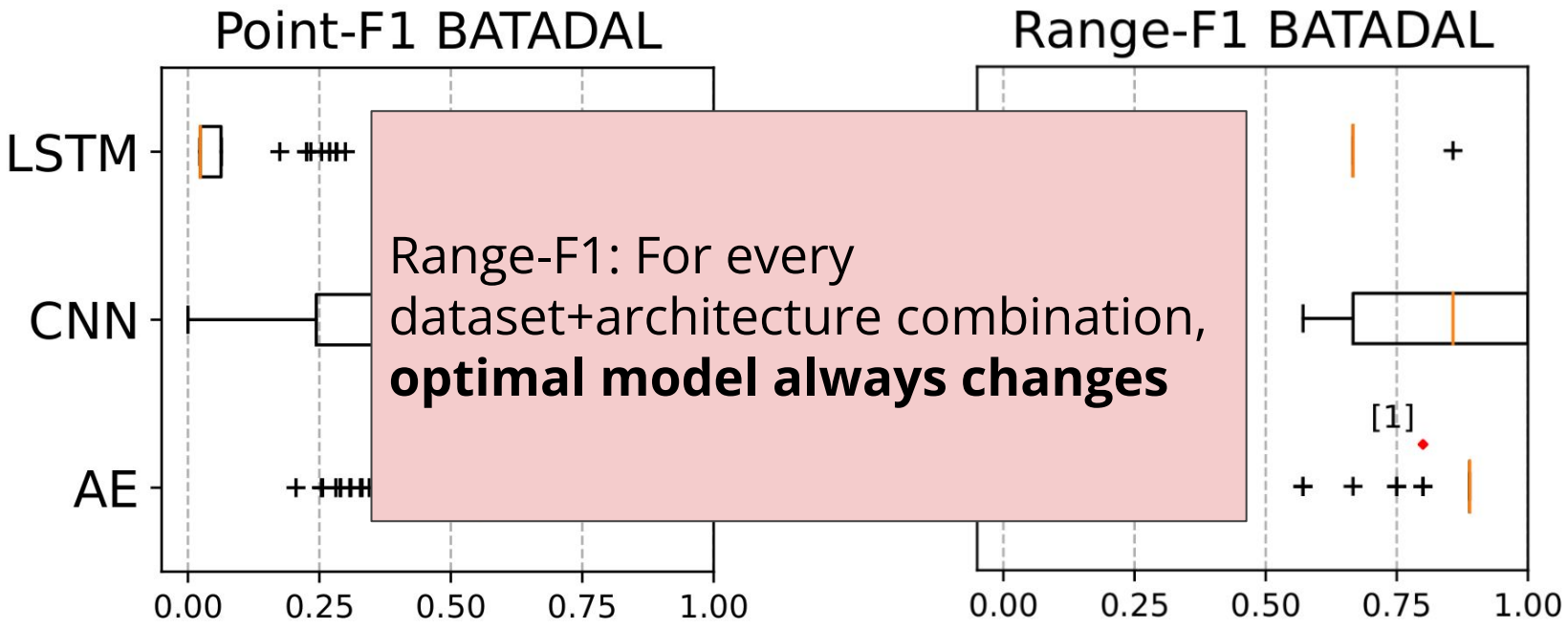
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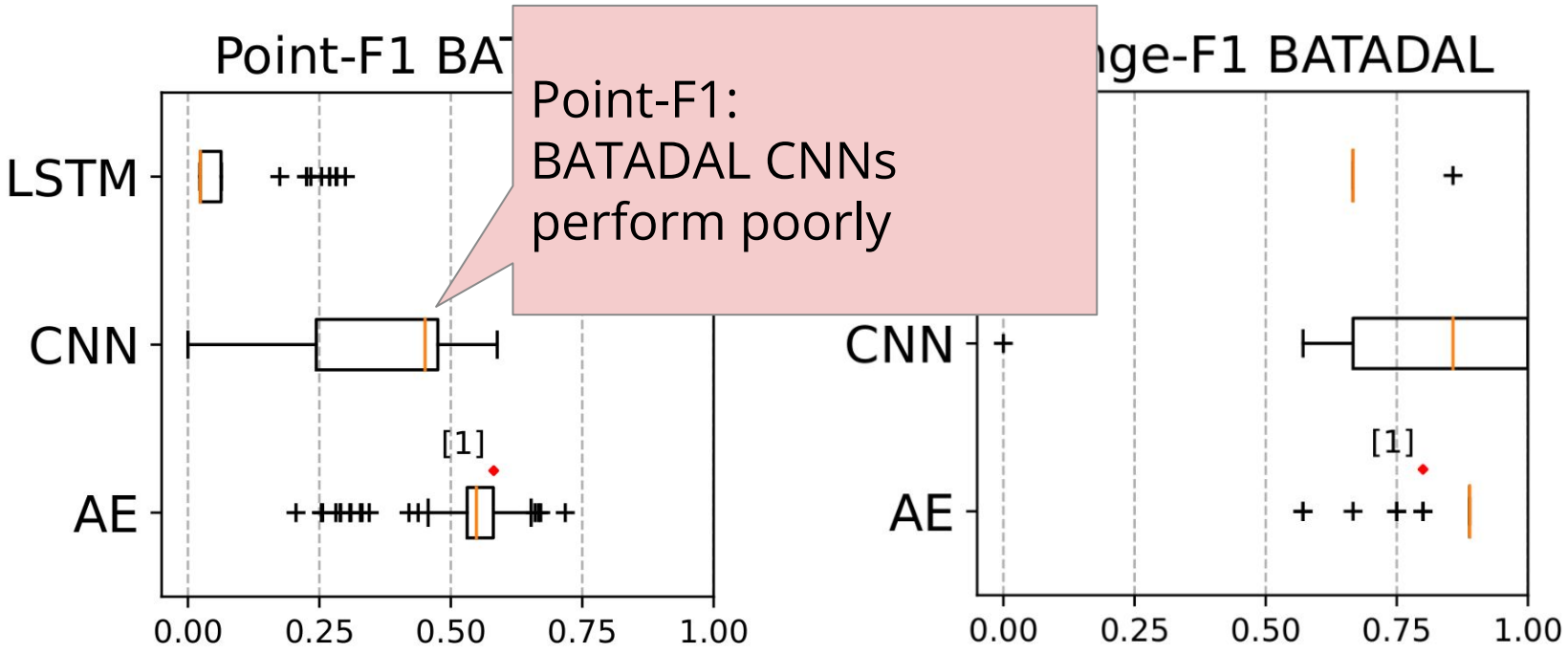
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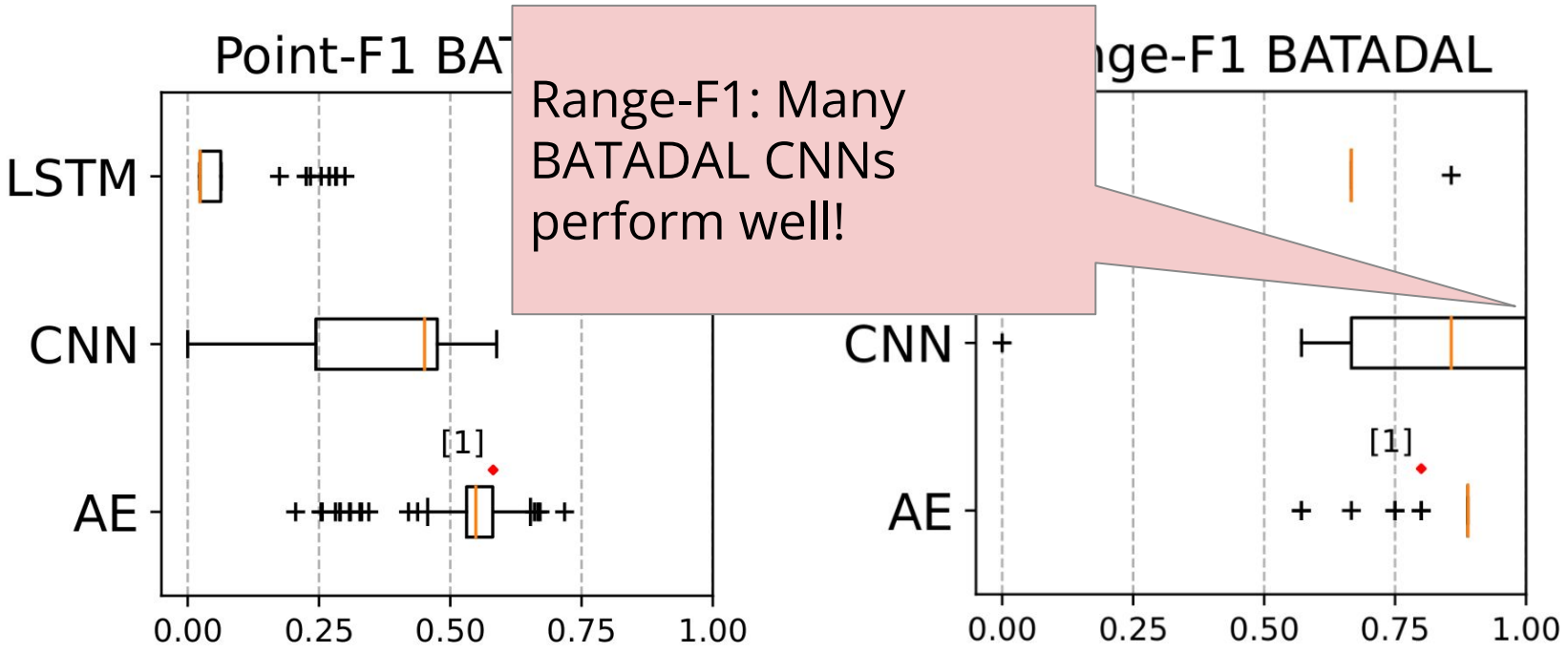
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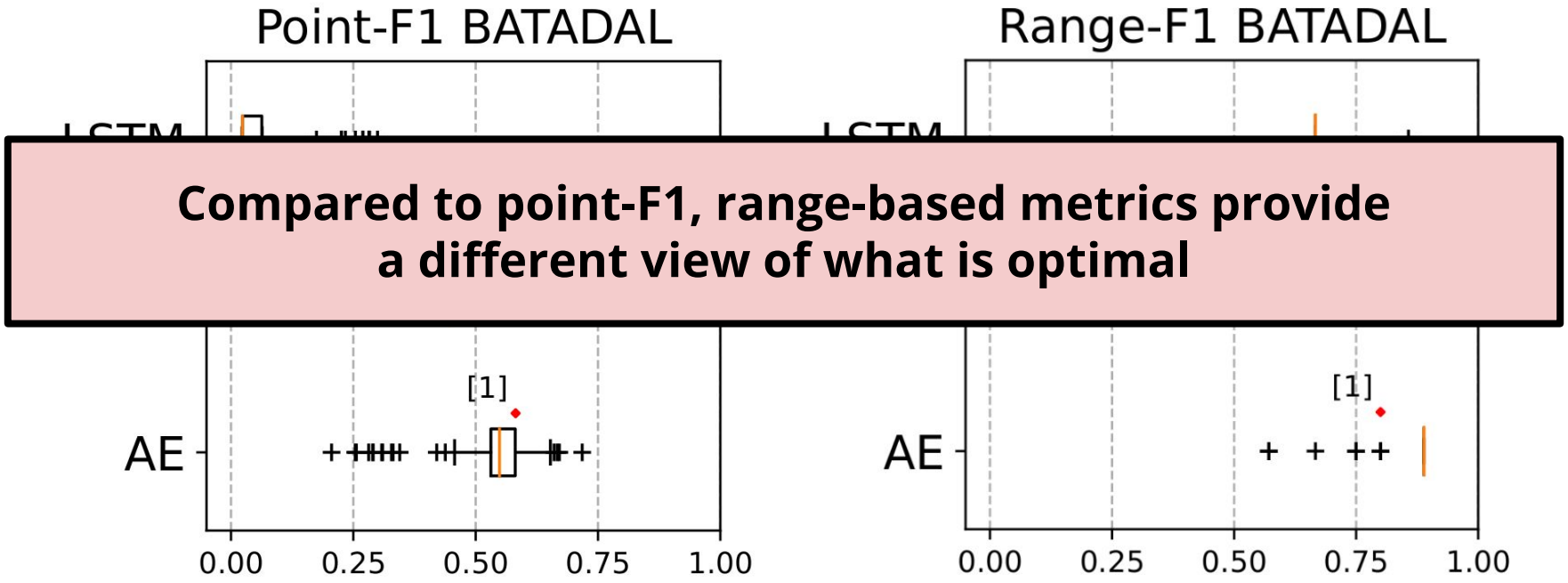
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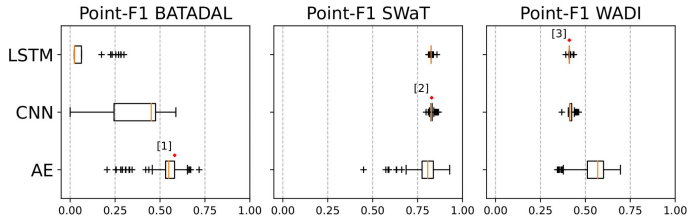

**Compared to point-F1, range-based metrics provide a different view of what is optimal**

# In summary

- We ask: **what are the best models** for ICS anomaly detection?
    - Establish a common methodology for **fair comparison**

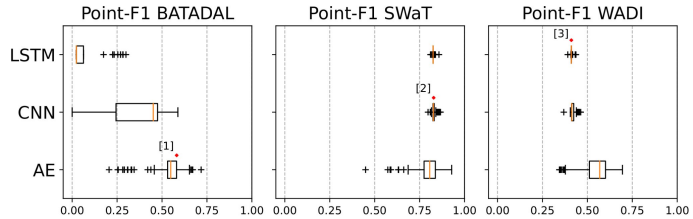
# In summary

- We ask: **what are the best models** for ICS anomaly detection?
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  - Identify four key preprocessing/training techniques
  - Find that **small models** are often just as effective!



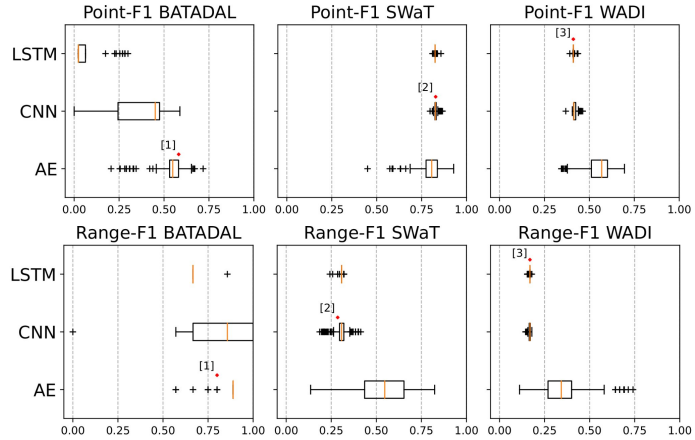
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  - Identify four key preprocessing/training techniques
  - Find that **small models** are often just as effective!
- **Point-F1 unfit** for time-series data
- **Range-based metrics** better measure usefulness of ICS anomaly detection



# In summary

- We ask: **what are the best models** for ICS anomaly detection?
  - Establish a common methodology for **fair comparison**
  - Identify four key preprocessing/training techniques
  - Find that **small models** are often just as effective!
- **Point-F1 unfit** for time-series data
- **Range-based metrics** better measure usefulness of ICS anomaly detection

**Perspectives from a Comprehensive Evaluation of Reconstruction-based Anomaly Detection in ICS**

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